



SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (AUTONOMOUS)

(Affiliated to JNTUK, Kakinada), (Recognized by AICTE, New Delhi)

Accredited by NAAC with 'A' Grade

CHINNA AMIRAM (P.O):: BHIMAVARAM :: W.G.Dt., A.P., INDIA :: PIN: 534 204

SCHEME OF INSTRUCTION & EXAMINATION

(Regulation R19)

M.TECH (INFORMATION TECHNOLOGY)

DEPARTMENT OF INFORMATION TECHNOLOGY

(With effect from 2019-2020 Admitted Batch onwards)

I-SEMESTER

Subject Code.	Name of the Subject	Category.	C	L	T	P	Internal Marks	External Marks	Total Marks
M19 IT 1101	Discrete Mathematical Structures	PC	3	3	0	0	25	75	100
M19 IT 1102	Advanced Data Structures	PC	3	3	0	0	25	75	100
#PE-I	Program Elective-I	PE	3	3	0	0	25	75	100
#PE-II	Program Elective-II	PE	3	3	0	0	25	75	100
M19 RD 1101	Research Methodology and IPR	CC	2	2	0	0	25	75	100
M19 IT 1113	Advanced Data Structures Lab	PC	2	0	0	4	25	75	100
M19 IT 1114	Computing Lab	PC	2	0	0	4	25	75	100
#AC-1	Audit course -1	AC	0	2	0	0	0	0	0
Total			18	16	0	8	175	525	700

	Code	Course		Code	Course
#PE-I	M19IT1103	Artificial Intelligence	# AC 1 & 2	M19AC0001	English for Research Paper Writing
	M19IT1104	Service Oriented Architectures And Web Security		M19AC0002	Disaster Management
	M19IT1105	Internet of Things		M19AC0003	Sanskrit for Technical Knowledge
	M19IT1106	Optimization Techniques		M19AC0004	Value Education
	M19IT1107	Parallel Computer Architecture		M19AC0005	Constitution of India
	M19IT1108	Bigdata Analytics		M19AC0006	Pedagogy Studies
#PE-II	M19IT1109	Principles Of Cryptography		M19AC0007	Stress Management by Yoga
	M19IT1110	Cluster And Grid Computing		M19AC0008	Personality Development through Life Enlightenment Skills.
	M19IT1111	Imaging and Multimedia Systems			
	M19IT1112	Advanced Graph Theory			

Code	Category	L	T	P	C	I.M	E.M	Exam
M19IT1101	PC	3	0	0	3	25	75	3 Hrs.
DISCRETE MATHEMATICAL STRUCTURES								
(Program Core-1)								
Course Objectives:								
1.	To understand the mathematical fundamentals that is prerequisites for a variety of courses like Data mining, Network protocols, analysis of Web traffic, Computer security, Software engineering, Computer architecture, operating systems, distributed systems, Bioinformatics, Machine learning.							
2.	To develop the understanding of mathematical and logical basis to many modern techniques in information technology like machine learning, programming language design and concurrency.							
3.	To study various sampling and classification problems.							
Course Outcomes: After completion of course, students will be able to:								
S.No	Outcome							Knowledge Level
1.	Utilize basic concepts of discrete and continuous probability theory.							K3
2	Make use of parametric families of distributions in analyzing problems							K3
3	Apply least squares method to fit best suitable curve for a given data							K3
4	Identify population parameters using different methods of estimation							K3
5.	Utilize the concepts of graph theory and trees in solving problems.							K3
6	Make use of counting techniques to solve combinatorial enumerator problem. Also solve recurrence relations by the use of different methods							K3
SYLLABUS								
UNIT-I (10 Hrs)	Review on basic Probability concepts- sample space -Addition and Multiplication theorems- Bay’s theorem- Random variables (Discrete & Continuous). Probability mass, density and cumulative distribution functions(for discrete & continuous random variables)-Joint distribution function- discrete & continuous random variables-conditional distribution- Expected value, Variance- Conditional expectation- Central limit theorem, probabilistic inequalities- Chebychev’s inequality, Jenson’s inequality- Cauchy – Schwartz inequatility- Convergence in probability- Markov chains.							
UNIT-II (10 Hrs)	Parametric families of distributions: Binomial, Poisson, Normal, Uniform, exponential and Gamma distribution, derivation of basic characteristics, Hyper geomentric & multinomial distributions.							
UNIT-III (10 Hrs)	Statistical inference, introduction to multivariate Statistical models: Scatter diagram, fitting of straight line, parabola, power and exponential curves: ax^b, ab^x, ae^{bx} using method of least square approximation, multiple regression (linear equations)- principal component analysis. Random Samples: Sampling distributions of Estimators, efficient and unbiased estimation, point & interval estimator: Method of moments and maximum likelyhood estimator.							

UNIT-IV (10 Hrs)	Graph Theory: Isomorphism, Planar graphs, Euler's and Hamiltonian circuits or Euler's cycles: graph coloring- Chromatic number. Trees: Introduction- Trees- Labeled tree- some diagrams on directed and undirected trees, Review of Basic properties of a tree- Sequential representation of a binary tree.
UNIT-V (10 Hrs)	Counting Methods: Permutation & Combination with & without repetition: Principal inclusion & exclusion: Special technique to solve combinational enumerator problem. Recurrence relation: Order & Degree of recurrence relation, Solution of Linear homogeneous and non-homogeneous recurrence relation with constant coefficient using methods of generating function, characteristic roots & particular solution by method of undetermined coefficients.
Text Books:	
1.	Murray R. Spiegel, Probability & Statistics, Schaum's Outline Series McGraw-Hill.
2.	Rosen K.H., Discrete Mathematical Structure, Sixth Edition, McGraw-Hill- International Edition.
3.	Miller and Freund's, Probability & Statistics for Engineers, Eighth Edition, By Pearson.
4.	Swapna Kumar chakraborty and Bikash kanti sarkar, Discrete Mathematics, Oxford Higher Education.
Reference Books:	
1.	Alan Tucker, Applied Combinatorics, 6 th Edition, Wiley.
2.	C.L.Liu, Elements of Discrete Mathematics, 2 nd Edition, McGraw-Hill Book Company
3.	John Vince, Foundation Mathematics for Computer Science, Springer
4.	K. Trivedi, Probability and Statistics with reliability, Queuing and Computer Science Applications, 2 nd edition, John Wiley and Sons Ltd. Chichester, UK.
5.	M. Mitzenmacher and E. Upfal, Probability and Computing: Randomized Algorithms and Probabilistic Analysis, Cambridge University Press.

Code	Category	L	T	P	C	I.M	E.M	Exam
M19 IT 1102	PC	3	0	0	3	25	75	3 Hrs.
ADVANCED DATA STRUCTURES								
(Program Core-2)								
Course Objectives:								
1.	The student should be able to choose appropriate data structures, understand the ADT/libraries and use it to design algorithms for a specific problem.							
2.	Students should be able to understand the necessary mathematical abstraction to solve problems.							
3.	To familiarize students with advanced paradigms and data structure used to solve algorithmic problems.							
4.	Student should be able to come up with analysis of efficiency and proofs of correctness.							
Course Outcomes								
After completion of course, students would be able to:								
S.No	Outcome							Knowledge Level
1.	Analyze programming problem statements.							K4
2.	Apply divide and conquer strategy to searching using iterative and/or recursive solutions.							K3
3.	Analyze algorithms for text processing applications and develop algorithms for computational geometry problems.							K4
4.	Analyze applications using data structure algorithms.							K4
SYLLABUS								
UNIT-I (10 Hrs)	Dictionaries: Definition, Dictionary Abstract Data Type, Implementation of Dictionaries. Hashing: Review of Hashing, Hash Function, Collision Resolution Techniques in Hashing, Separate Chaining, Open Addressing, Linear Probing, Quadratic Probing, Double Hashing, Rehashing, Extendible Hashing.							
UNIT-II (10 Hrs)	Skip Lists: Need for Randomizing Data Structures and Algorithms, Search and Update Operations on Skip Lists, Probabilistic Analysis of Skip Lists, Deterministic Skip Lists							
UNIT-III (10 Hrs)	Trees: Binary Search Trees, AVL Trees, Red Black Trees, Trees, B-Trees, Splay Trees							
UNIT-IV (10 Hrs)	Text Processing: Sting Operations, Brute-Force Pattern Matching, The Boyer-Moore Algorithm, The Knuth-Morris-Pratt Algorithm, Standard Tries, Compressed Tries, Suffix Tries, The Huffman Coding Algorithm, The Longest Common Subsequence Problem (LCS), Applying Dynamic Programming to the LCS Problem.							

UNIT-V (10 Hrs)	Computational Geometry: One Dimensional Range Searching, Two Dimensional Range Searching, Constructing a Priority Search Tree, Searching a Priority Search Tree, Priority Range Trees, Quad trees, k- D Trees. Recent Trends in Hashing, Trees, and various computational geometry methods for efficiently solving the new evolving problem
Text Books:	
1.	Mark Allen Weiss, Data Structures and Algorithm Analysis in C++, 2nd Edition, Pearson, 2004.
2.	M T Goodrich, Roberto Tamassia, Algorithm Design, John Wiley, 2002.
Reference Books:	
1.	PETER BRASS , "Advanced Data Structures" , CAMBRIDGE UNIVERSITY PRESS, 2008.

Code	Category	L	T	P	C	I.M	E.M	Exam
M19 IT 1103	PE	3	0	0	3	25	75	3 Hrs.
ARTIFICIAL INTELLIGENCE								
(Program Elective-1)								
Course Objectives:								
1.	Gain a historical perspective of AI and its foundations.							
2.	Become familiar with basic principles of AI toward problem solving, inference, perception, knowledge representation, and learning.							
3.	Investigate applications of AI techniques in intelligent agents, expert systems, artificial neural networks and other machine learning models.							
4.	Experience AI development tools such as an „AI language“, expert system shell, and/or data mining tool. Experiment with a machine learning model for simulation and analysis.							
5.	Solve problems with uncertain information using Bayesian approaches.							
Course Outcomes At the end of the course, student will be able to								
S.No	Outcome							Knowledge Level
1.	Able to design & construct knowledge using representation tools & use of problem solving strategies for real time problem solving in terms intelligent agents							K3
2.	Attain the capability to represent various real life problem domains using logic based techniques and use this to perform inference or planning.							K3
3.	Able to apply various reasoning techniques to solve the uncertain & incomplete problems using probability theory							K3
4.	Able to solve problems by designing the ANN models for classification & prediction							K5
SYLLABUS								
UNIT-I (10 Hrs)	Introduction to artificial intelligence: Introduction , history, intelligent systems, foundations of AI, applications, tic-tac-tie game playing, development of AI languages, current trends in AI, Problem solving: state-space search and control strategies: Introduction, general problem solving, characteristics of problem, exhaustive searches, heuristic search techniques, iterative-deepening a*, constraint satisfaction							
UNIT-II (10 Hrs)	Problem reduction and game playing: Introduction, problem reduction, game playing, alpha-beta pruning, two-player perfect information games, Logic concepts: Introduction, propositional calculus, propositional logic, natural deduction system, axiomatic system, semantic tableau system in propositional logic, resolution refutation in propositional logic, predicate logic							
UNIT-III (10 Hrs)	Knowledge representation: Introduction, approaches to knowledge representation, knowledge representation using semantic network, extended semantic networks for KR, knowledge representation using frames, advanced knowledge representation techniques:							

	Introduction, conceptual dependency theory, script structure, cyc theory, case grammars, semantic web, Expert system and applications: Introduction phases in building expert systems, expert system versus traditional systems, rule-based expert systems blackboard systems truth maintenance systems, application of expert systems, list of shells and tools
UNIT-IV (10 Hrs)	Uncertainty measure: probability theory: Introduction, probability theory, Bayesian belief networks, certainty factor theory, dempster-shafer theory , Fuzzy sets and fuzzy logic: Introduction, fuzzy sets, fuzzy set operations, types of membership functions, multi valued logic, fuzzy logic, linguistic variables and hedges, fuzzy propositions, inference rules for fuzzy propositions, fuzzy systems.
UNIT-V (10 Hrs)	Machine learning paradigms: Introduction, machine learning systems, supervised and unsupervised learnings, inductive learning, deductive learning, clustering, support vector machines, case based reasoning and learning, Artificial neural networks: Introduction, artificial networks, single layer feed forward networks, multi layered forward networks, design issues of artificial neural networks
Text Books:	
1.	Artificial Intelligence- Saroj Kaushik, CENGAGE Learning,
2.	Artificial intelligence, A modern Approach, 2nd ed, Stuart Russel, Peter Norvig, PEA
3.	Artificial Intelligence- 3rd ed, Rich, Kevin Knight, Shiv Shankar B Nair, TMH
4.	Introduction to Artificial Intelligence, Patterson, PHI
Reference Books:	
1.	Artificial intelligence, structures and Strategies for Complex problem solving, 5th ed, George F Luger, PEA
2.	Introduction to Artificial Intelligence, Ertel, Wolf Gang, Springer
3.	Artificial Intelligence, A new Synthesis, Nils J Nilsson, Elsevier

Code	Category	L	T	P	C	I.M	E.M	Exam
M19 IT 1104	PE	3	0	0	3	25	75	3 Hrs.
SERVICE ORIENTED ARCHITECTURES AND WEB SECURITY								
(Program Elective-1)								
Course Objectives:								
1.	To provide an overview of XML Technology and modeling databases inXML							
2.	The students will learn the concepts of SOA and Web services, some of the prevailing standards and technologies of WebServices							
3.	The students will also learn the approaches for providing security for XML documents as well as messages exchanged among WebServices							
Course Outcomes By the end of this course student will be able to								
S.No	Outcome							Knowledge Level
1.	Demonstrate XML Technology and modelling databases in XML							K2
2.	Analyze SOA and Web services, some of the prevailing							K4
3.	Analyze standards and technologies of Web Services and Security Frame Works							K4
SYLLABUS								
UNIT-I (10 Hrs)	XML Technology: XML – XML and Web - Name Spaces – XML Document Structure - Structuring with Schemas and DTD - Modeling Databases in XML – XQuery							
UNIT-II (10 Hrs)	SOA Basics : Service Oriented Architecture (SOA) – Comparing SOA with Client-Server and Distributed architectures - Characteristics of SOA – Benefits of SOA -- Principles of Service orientation – Service layers - Business Process management							
UNIT-III (10 Hrs)	Web Services (WS) : SOA and Web Services – Web Services Protocol Stack – Service descriptions – WSDL – Messaging with SOAP – Service discovery – UDDI. Service-Level Interaction patterns – XML and Web Services - Enterprise Service Bus - .NET and J2EE Interoperability.							
UNIT-IV (10 Hrs)	WS Technologies and Standards :Web Services Technologies - JAX-RPC, JAX-WS. Web Service Standards – WS-RM, WS-Addressing, WS-Policy. Service Orchestration and Choreography – Composition Standards - BPEL. Service Oriented Analysis and Design.							
UNIT-V (10 Hrs)	XML and WS Security :XML Security Overview – Canonicalization – XML Security Framework – XML Encryption – XML Signature – XKMS Structure. Web Services Security - XACML - WS-Security.							
Text Books:								
1.	Ron Schmelzer et al. “XML and Web Services”, Pearson Education, 2008. (Unit 1 and3)							

2.	Thomas Erl, “ Service Oriented Architecture: Concepts, Technology, and Design”, Pearson Education, 2005 (Unit 2, 3, 4, and5)
3.	Frank P.Coyle, “XML, Web Services and the Data Revolution”, Pearson Education, 2002 (Unit 5)
Reference Books:	
1.	Eric Newcomer, Greg Lomow, “Understanding SOA with Web Services”, Addison Wesley, 2005.
2.	James McGovern, Sameer Tyagi, Michael E Stevens, Sunil Mathew, “Java Web Services Architecture”, Elsevier,2011.
3.	Mark O’ Neill, et al., “Web Services Security”, Tata McGraw-Hill Edition,2003.
4.	Sandeep Chatterjee and James Webber, “Developing Enterprise Web Services: An Architect's Guide”, Prentice Hall, 2004

Code	Category	L	T	P	C	I.M	E.M	Exam
M19IT1105	PE	3	0	0	3	25	75	3 Hrs.
INTERNET OF THINGS								
(Program Elective-1)								
Course Objectives:								
1.	To Understand Smart Objects and IoT Architectures.							
2.	To learn about various IOT-related protocols.							
3.	To build simple IoT Systems using Arduino and Raspberry Pi.							
4.	To understand data analytics and cloud in the context of IoT.							
5.	To develop IoT infrastructure for popular applications.							
Course Outcomes								
S.No	Outcome							Knowledge Level
1.	Evaluate the concept of 'internet of things' in different contexts.							K4
2.	Analyze various protocols for IoT.							K4
3.	Design a PoC of an IoT system using Rasperry Pi/Arduino.							K6
4.	Apply data analytics and use cloud offerings related to IoT.							K3
5.	Analyze applications of IoT in real time scenario.							K4
SYLLABUS								
UNIT-I (10 Hrs)	FUNDAMENTALS OF IoT: Evolution of Internet of Things, Enabling Technologies, IoT Architectures,oneM2M, IoT World Forum (IoTWF) and Alternative IoT models, Simplified IoT Architecture and Core IoT Functional Stack, Fog, Edge and Cloud in IoT, Functional blocks of an IoT ecosystem, Sensors, Actuators, Smart Objects and Connecting Smart Objects.							
UNIT-II (10 Hrs)	IoT PROTOCOLS: IT Access Technologies: Physical and MAC layers, topology and Security of IEEE 802.15.4, 802.15.4g, 802.15.4e, 1901.2a, 802.11ah and Lora WAN, Network Layer: IP versions, Constrained Nodes and Constrained Networks, Optimizing IP for IoT: From 6LoWPAN to 6Lo, Routing over Low Power and Lossy Networks, Application Transport Methods: Supervisory Control and Data Acquisition, Application Layer Protocols: CoAP and MQTT.							
UNIT-III (10 Hrs)	DESIGN AND DEVELOPMENT: Design Methodology, Embedded computing logic, Microcontroller, System on Chips, IoT system building blocks, Arduino, Board details, IDE programming, Raspberry Pi, Interfaces and Raspberry Pi with Python Programming.							
UNIT-IV (10 Hrs)	DATA ANALYTICS AND SUPPORTING SERVICES: Structured Vs Unstructured Data and Data in Motion Vs Data in Rest, Role of Machine Learning – No SQL Databases, Hadoop Ecosystem, Apache Kafka, Apache Spark, Edge Streaming Analytics and Network Analytics, Xively Cloud for IoT, Python Web Application Framework, Django, AWS for IoT, System Management with NETCONF-YANG.							

UNIT-V (8 Hrs)	CASE STUDIES/INDUSTRIAL APPLICATIONS: Cisco IoT system, IBM Watson IoT platform, Manufacturing, Converged Plant wide Ethernet Model (CPwE), Power Utility Industry, Grid Blocks Reference Model, Smart and Connected Cities: Layered architecture, Smart Lighting, Smart Parking Architecture and Smart Traffic Control.
Text Books:	
1.	IoT Fundamentals: Networking Technologies, Protocols and Use Cases for Internet of Things, David Hanes, Gonzalo Salgueiro, Patrick Grossetete, Rob Barton and Jerome Henry, Cisco Press, 2017
Reference Books:	
1.	Internet of Things – A hands-on approach, Arshdeep Bahga, Vijay Madisetti, Universities Press, 2015.

Code	Category	L	T	P	C	I.M	E.M	Exam
M19 IT 1106	PE	3	0	0	3	25	75	3 Hrs.
OPTIMIZATION TECHNIQUES								
(Program Elective-1)								
Course Objectives:								
1.	To understand the theory of optimization methods and algorithms developed for solving various types of optimization problems.							
2.	To develop and promote research interest in applying optimization techniques in problems of Engineering and Technology.							
3.	To apply the mathematical results and numerical techniques of optimization theory to concrete Engineering problems							
Course Outcomes At the end of the course, the student should be able to								
S.No	Outcome							Knowledge Level
1.	Students should able to Apply the dynamic programming to solve problems of discreet and continuous variables.							K3
2.	Students should able to Apply the concept of non-linear programming							K3
3.	Students should able to Analyze out sensitivity							K4
4.	Student should able to Analyze the real world problem and simulate it.							K4
SYLLABUS								
UNIT-I (10 Hrs)	Optimization Techniques, Model Formulation, models, General L.R Formulation, Simplex Techniques, Sensitivity Analysis, Inventory Control Models							
UNIT-II (10 Hrs)	Formulation of a LPP - Graphical solution revised simplex method - duality theory - dual simplex method - sensitivity analysis - parametric programming							
UNIT-III (10 Hrs)	Nonlinear programming problem - Kuhn-Tucker conditions min cost flow problem - max flow problem - CPM/PERT							
UNIT-IV (10 Hrs)	Scheduling and sequencing - single server and multiple server models - deterministic inventory models - Probabilistic inventory control models - Geometric Programming.							
UNIT-V (10 Hrs)	Competitive Models,Single and Multi-channel Problems, Sequencing Models, Dynamic Programming, Flow in Networks, Elementary Graph Theory, Game Theory Simulation							
Text Books:								
1.	H.A. Taha, Operations Research, An Introduction, PHI, 2008							
2.	H.M. Wagner, Principles of Operations Research, PHI, Delhi, 1982.							

Reference Books:

1.	J.C. Pant, Introduction to Optimisation: Operations Research, Jain Brothers, Delhi, 2008
2.	Hitler Libermann Operations Research: McGraw Hill Pub. 2009
3.	Pannerselvam, Operations Research: Prentice Hall of India 2010
4.	Harvey M Wagner, Principles of Operations Research: Prentice Hall of India 2010

Code	Category	L	T	P	C	I.M	E.M	Exam
M19 IT 1107	PE	3	0	0	3	25	75	3 Hrs.
PARALLEL COMPUTER ARCHITECTURE								
(Program Elective-1)								
Course Objectives:								
1.	To understand the principles of parallel computer architecture							
2.	To understand the design of parallel computer systems including modern parallel architectures							
3.	To assess the communication and computing possibilities of parallel system architecture and to predict the performance of parallel applications							
Course Outcomes								
After completion of course, students would be able to:								
S.No	Outcome							Knowledge Level
1.	Assume representation of data, addressing modes, and instructions sets.							K4
2.	Identify parallelism both in terms of a single processor and multiple processors							K3
3.	Analyze parallel hardware constructs to include instruction-level parallelism for multi core processor design							K4
SYLLABUS								
UNIT-I (10 Hrs)	Fundamentals of Computer Design: Defining Computer Architecture – Trends in Technology – Trends in Power in Integrated Circuits– Trends in Cost – Dependability – Measuring, Reporting and Summarizing Performance – Quantitative Principles of Computer Design – Basic and Intermediate concepts of pipelining – Pipeline Hazards – Pipelining Implementation issues.							
UNIT-II (10 Hrs)	Instruction-Level Parallelism and Its Exploitation: Instruction-Level Parallelism: Concepts and Challenges – Basic Compiler Techniques for Exposing ILP – Reducing Branch Costs with Prediction – Overcoming Data Hazards with Dynamic Scheduling – Dynamic Scheduling: Algorithm and Examples – Hardware-Based Speculation – Exploiting ILP Using Multiple Issue and Static Scheduling – Exploiting ILP Using Dynamic Scheduling, Multiple Issue and Speculation – Studies of the Limitations of ILP – Limitations on ILP for Realizable Processors – Hardware versus Software Speculation – Using ILP Support to Exploit Thread-Level Parallelism							
UNIT-III (10 Hrs)	Data-Level and Thread-Level Parallelism: Vector Architecture – SIMD Instruction Set Extensions for Multimedia – Graphics Processing Units – Detecting and Enhancing Loop-Level Parallelism – Centralized Shared-Memory Architectures – Performance of Shared-Memory Multiprocessors – Distributed Shared Memory and Directory Based Coherence – Basics of Synchronization – Models of Memory Consistency – Programming Models and Workloads for Warehouse-Scale Computers – Computer Architecture of Warehouse-Scale Computers – Physical Infrastructure and Costs of Warehouse-Scale Computers							

UNIT-IV (10 Hrs)	Memory Hierarchy Design: Cache Performance – Six Basic Cache Optimizations – Virtual Memory – Protection and Examples of Virtual Memory – Ten Advanced Optimizations of Cache Performance – Memory Technology and Optimizations – Protection: Virtual Memory and Virtual Machines – The Design of Memory Hierarchies
UNIT-V (10 Hrs)	Storage Systems & Case Studies: Advanced Topics in Disk Storage – Definition and Examples of Real Faults and Failures – I/O Performance, Reliability Measures and Benchmarks – Designing and Evaluating an I/O System – The Internet Archive Cluster Case Studies / Lab Exercises: INTEL i3, i5, i7 processor cores, NVIDIA GPUs, AMD, ARM processor cores – Simulators – GEM5, CACTI, SIMICS, Multi2sim and INTEL Software development tools.
Text Books:	
1.	David.A.Patterson, John L.Hennessy, "Computer Architecture: A Quantitative approach", Elsevier, 5 th Edition 2012.
2.	K.Hwang, Naresh Jotwani, “Advanced Computer Architecture, Parallelism, Scalability, Programmability”, Tata McGraw Hill, 2 nd Edition 2010
Reference Books:	
1	David E. Culler & Jaswinder Pal Singh, “Parallel Computing Architecture: A Hardware / Software Approach”, Morgan Kaufman Publishers, 1999.
2	Michael J. Quinn, “Parallel Programming in C with MPI & OpenMP”, Tata McGraw-Hill, New Delhi, 2003.

Code	Category	L	T	P	C	I.M	E.M	Exam
M19 IT 1108	PE	3	0	0	3	25	75	3 Hrs.
BIG DATA ANALYTICS								
(Program Elective-2)								
Course Objectives:								
1.	To Provide an overview of an exciting growing field of Big Data analytics.							
2.	To Introduce the tools required to manage and analyze big data like Hadoop, MapReduce etc.,							
Course Outcomes								
S.No	Outcome							Knowledge Level
1.	Identify the programming requirements viz., generic types and methods to perform data analysis							K3
2.	Apply the existing technologies in distributed files systems to analyze the big data							K3
3.	To analyze the Map-Reduce programming model for better optimization							K4
4.	Collect, manage, store, query, and analyze big data; and identify the need of interfaces to perform I/O operations in Hadoop							K3
SYLLABUS								
UNIT-I (10 Hrs)	Data structures in Java: Linked List, Stacks, Queues, Sets, Maps; Generics: Generic classes and Type parameters, Implementing Generic Types, Generic Methods, Wrapper Classes, Concept of Serialization							
UNIT-II (10 Hrs)	Working with Big Data: Google File System, Hadoop Distributed File System (HDFS), Building blocks of Hadoop (Name node, Data node, Secondary Name node, Job Tracker, Task Tracker), Introducing and Configuring Hadoop cluster (Local, Pseudo-distributed mode, Fully Distributed mode), Configuring XML files.							
UNIT-III (10 Hrs)	Writing Map Reduce Programs: A Weather Dataset, Understanding Hadoop API for Map Reduce Framework (Old and New), Basic programs of Hadoop Map Reduce: Driver code, Mapper code, Reducer code, Record Reader, Combiner, Partitioner							
UNIT-IV (10 Hrs)	Hadoop I/O: The Writable Interface, Writable Comparable and comparators, Writable Classes: Writable wrappers for Java primitives, Text, Bytes Writable, Null Writable, Object Writable and Generic Writable, Writable collections, Implementing a Custom Writable: Implementing a Raw Comparator for speed, Custom comparators							
UNIT-V (10 Hrs)	Pig: Hadoop Programming Made Easier, Admiring the Pig Architecture, Going with the Pig Latin Application Flow, Working through the ABCs of Pig Latin, Evaluating Local and Distributed Modes of Running Pig Scripts, Checking out the Pig Script Interfaces, Scripting with Pig Latin, Applying Structure to Hadoop Data with Hive: Saying Hello to Hive, Seeing How							

	the Hive is Put Together, Getting Started with Apache Hive, Examining the Hive Clients, Working with Hive Data Types, Creating and Managing Databases and Tables, Seeing How the Hive Data Manipulation Language Works, Querying and Analyzing Data
Text Books:	
1.	Big Java 4th Edition, Cay Horstmann, Wiley John Wiley & Sons, INC
2.	Hadoop: The Definitive Guide by Tom White, 3rd Edition, O'Reilly
3.	Hadoop in Action by Chuck Lam, MANNING Publ
4.	Hadoop for Dummies by Dirk deRoos, Paul C.Zikopoulos, Roman B.Melnyk, Bruce Brown and Rafael Coss
Reference Books:	
1.	Hadoop in Practice by Alex Holmes, MANNING Publ
2.	Hadoop MapReduce Cookbook, Srinath Perera, Thilina Gunarathne
Web Resources:	
1.	Hadoop: https://hadoop.apache.org/
2.	Hive: https://cwiki.apache.org/confluence/display/Hive/Home/
3.	Piglatin: https://pig.apache.org/docs/r0.7.0/tutorial.html

Code	Category	L	T	P	C	I.M	E.M	Exam
M19 IT 1109	PE	3	0	0	3	25	75	3 Hrs.
PRINCIPLES OF CRYPTOGRAPHY								
(Program Elective-2)								
Course Objectives:								
1.	To gain knowledge about the mathematics of the cryptographic algorithms.							
2.	To get an insight into the working of different existing cryptographic algorithms.							
3.	Analyzing and application of cryptography for secure eCommerce and other secret transactions							
Course Outcomes								
S.No	Outcome							Knowledge Level
1.	Identify classical cryptosystems to security problems							K3
2.	Apply the existing cryptographic algorithms with the existing communication protocols							K3
3.	Analyzing the application of cryptography for secure eCommerce and other secret transactions							K4
SYLLABUS								
UNIT-I (10 Hrs)	Algebra: Group, cyclic group, cyclic subgroup, field, probability. Number Theory: Fermat's theorem , Cauchy 's theorem, Chinese remainder theorem, primality testing algorithm, Euclid's algorithm for integers, quadratic residues, Legendre symbol, Jacobi symbol etc..							
UNIT-II (10 Hrs)	Cryptography and cryptanalysis, Classical Cryptography, substitution cipher, different type of attack: CMA,CPA,CCA etc, Shannon perfect secrecy, OTP, Pseudo random bit generators, stream ciphers and RC4.							
UNIT-III (10 Hrs)	Block ciphers: Modes of operation, DES and its variants, AES, linear and differential cryptanalysis.							
UNIT-IV (10 Hrs)	One-way function , trapdoor one-way function, Public key cryptography, RSA cryptosystem, DiffieHellman key exchange algorithm, Elgamal Cryptosystem.							
UNIT-V (10 Hrs)	Cryptographic hash functions, secure hash algorithm, Message authentication, digital signature, RSA digital signature, Elgamal digital signature.							
Text Books:								
1.	Stinson. D. Cryptography: Theory and Practice, third edition, Chapman & Hall/CRC, 2010.							
Reference Books:								
1.	W. Stallings, Cryptography and Network Security Principles and practice, 5/e, Pearson Education Asia, 2012.							
2.	Behrouz A. Forouzan and Debdeep Mukhopadhyay, Cryptography and Network Security, second edition, Tata McGraw Hill, 2011							
3.	Thomas Koshy, Elementary Number Theory with applications, Elsevier India, 2005.							

Code	Category	L	T	P	C	I.M	E.M	Exam
M19 IT 1110	PE	3	0	0	3	25	75	3 Hrs.
CLUSTER AND GRID COMPUTING								
(Program Elective-2)								
Course Objectives:								
1.	To provide an insight for achieving cost efficient high performance system.							
2.	To deal with design and architecture of grid and cluster computing.							
3.	To able to setup various clustering techniques in parallel computing and applying various process scheduling policies.							
Course Outcomes At the end of the course, the student should be able to								
S.No	Outcome							Knowledge Level
1.	Identify the Functionalities & Applications of Grid Computing, Web Services, and Service-oriented architecture.							K3
2	Apply and Analyze Resource management and Scheduling in Grid computing & Cluster Computing.							K4
3	Develop how to setup a cluster, Apply various Process Scheduling Algorithms and load balancing.							K3
SYLLABUS								
UNIT-I (10 Hrs)	Introduction: Cluster and Grid computing, Meta-computing, Web services and Grid Computing, e-Governance and the Grid Technologies and Architectures for Grid Computing: Issues in Data Grids, Functional requirements in Grid Computing, Standards for Grid Computing, Recent technology trends in Large Data Grids. Web Services and the Service Oriented Architecture: Service Oriented Architecture, SOAP and WSDL, Creating Web Services, Server Side.							
UNIT-II (10 Hrs)	OGSA and WSRF: OGSA for Resource Distribution, Stateful Web Services in OGSA, WSRF, WSRF Specification, Globus Toolkit: History, version, Applications, Approaches and Benefits, Infrastructure Management, Monitoring and Discovery, Security, Data Choreography and Coordination, GT4 Architecture, GT4 Containers. The Grid and Databases: Requirements, Storage Request Broker, Integration of Databases with the Grid, Architecture of OGSADAI for offering Grid Database services.							
UNIT-III (10 Hrs)	Cluster Computing: Approaches to Parallel Computing, Definition and Architecture of a Cluster, Categories of clusters. Cluster Middleware: Levels and Layers of Single System Image, Design objectives, Resource Management and Scheduling, Cluster programming Environment and Tools. Networking, Protocols & I/O for clusters: Networking and Interconnection/Switching Devices, Design Issues, Design Architecture, HiPPI, ATM, Myrinet, Memory Channel							
UNIT-IV (10 Hrs)	Setting Up and Administering a Cluster: Setup of simple cluster, setting up nodes, clusters of clusters, System monitoring, Global Clocks Sync. Cluster Technology for High Availability: High availability clusters, high availability parallel computing, types of							

	failures and errors, cluster architectures and configurations for high availability, Failure/Recovery clusters.
UNIT-V (10 Hrs)	Process Scheduling: Job management System, Resource management system, policies of resource utilization, Scheduling policies. Load Sharing and Load Balancing: Introduction, Strategies for load balancing, Modelling parameters. Recent trends: technologies and attributes in Cluster and Grid computing. Case study of various cluster architectures, load balancing and scheduling policies.
Text Books:	
1.	Grid and Cluster Computing , CSR Prabhu, PHI
Reference Books:	
1.	Distributed and Cloud Computing: From Parallel Processing to the Internet of Things, K. Hwang, G. Fox and J. Dongarra, MK Publishers

Code	Category	L	T	P	C	I.M	E.M	Exam
M19 IT 1111	PE	3	0	0	3	25	75	3 Hrs.
IMAGING AND MULTIMEDIA SYSTEMS								
(Program Elective-2)								
Course Objectives:								
1.	To understand the basics of image processing and image security techniques							
2.	To study various compression and file formats used in imaging and multimedia systems							
3.	To analyze different media and design issues related to multimedia systems							
Course Outcomes								
S.No	Outcome							Knowledge Level
1.	Able to design the models for new compression standards							K3
2.	Acquire skill set to handle all multimedia components & able to save the images in various formats							K2
3.	Able to develop Integrated and Collaborative multimedia systems							K3
4.	Able to apply various information security methods like stenography							K3
5.	Able to design an effective user interface using HTML & XML							K4
SYLLABUS								
UNIT-I (10 Hrs)	Introduction to Image Processing: Steps in Image Processing Systems –Image Acquisition – Sampling and Quantization – Pixel Relationships – Colour Fundamentals and Models. Introduction to Multimedia: Multimedia Elements – Multimedia applications – Multimedia System Architecture – Evolving technologies for Multimedia – Defining objects for Multimedia systems – Multimedia Data interface standards – Multimedia Databases.							
UNIT-II (10 Hrs)	Compression and Decompression: Need for Data Compression – Types of Compression – Binary Image Compression Schemes – Image Compression – Video Compression – Audio Compression. Data and File Format Standards: Rich Text Format – TIFF File Format – Resource Interface File Format – MIDI File Format - JPEG DIB File Format – AVI Indeo File Format – MPEG Standards							
UNIT-III (10 Hrs)	Image computing: The basics of processing 2D images- Thresholding -Convolution-Edge detection-Mathematical Morphology and Shape Descriptors-Noise Reduction- Image Fusion. Image Security: Image Forensics - Steganography -Image Cryptography Techniques-Chaos based and Non-Chaos based methods.							
UNIT-IV (10 Hrs)	Input and Output Technologies: Multimedia I/O Technologies: Image Scanners – Digital Voice and Audio – Digital Camera – Video Images and Animation – Full Motion Video - Video Motion Analysis.							

UNIT-V (10 Hrs)	Multimedia Application Classes – Types of Multimedia Systems – Virtual Reality – Components of Multimedia Systems -Multimedia Authoring Systems – Multimedia Authoring Tools - User Interface Design- Mobile Messaging – Hypermedia Message Components -Hypermedia Linking and embedding.
Text Books:	
1.	Rafael C Gonzalez, Richard E Woods 2nd Edition, Digital Image Processing - Pearson Education, 2011.
2.	Ralf Steinmetz, Klara Steinmetz, “Multimedia Computing, Communications & Applications”, Pearson education, 2009.
Reference Books:	
1.	A.K. Jain, Fundamentals of Digital Image Processing ,PHI, New Delhi, 2001.
2.	William K Pratt, Digital Image Processing, John Willey , 2012.
3.	Prabat K Andleigh and Kiran Thakrar, “Multimedia Systems and Design”, Prentice Hall India, 2007,New Delhi.
4.	Tay Vaughan, “Multimedia Making It Work”, McGraw Hill, 2011. 5. Parekh R “Principles of Multimedia” Tata McGraw-Hill, 2006.

Code	Category	L	T	P	C	I.M	E.M	Exam
M19 IT 1112	PE	3	0	0	3	25	75	3 Hrs.
ADVANCED GRAPH THEORY								
(Program Elective-2)								
Course Objectives:								
1.	All elementary concepts such as coloring, covering, hamiltonicity, planarity, connectivity and so on, it will also introduce the students to some advanced concepts.							
2.	The student will know the definitions of relevant vocabulary and various algorithms from graph theory.							
Course Outcomes								
S.No	Outcome							Knowledge Level
1.	Evaluate precise and accurate mathematical definitions of objects in graph theory.							K5
2.	Analyze and solve some real time problems using concepts of graph theory (e.g., scheduling problems).							K4
3.	Develop some classical graph algorithms in order to find sub graphs with desirable properties.							K3
SYLLABUS								
UNIT-I (10 Hrs)	Basic Concepts- Graphs and digraphs, incidence and adjacency matrices, isomorphism, the automorphism group, Trees- Equivalent definitions of trees and forests, Cayley's formula, the Matrix-Tree theorem							
UNIT-II (10 Hrs)	Connectivity- Cut vertices, cut edges, bonds, the cycle space and the bond space, blocks, Menger's theorem, Paths and Cycles- Euler tours, Hamilton paths and cycles, theorems of Dirac, Ore, Bondy and Chvatal, circumference, the Chinese Postman Problem, the Travelling Salesman problem, diameter and maximum degree.							
UNIT-III (10 Hrs)	Matchings- Berge's Theorem, perfect matchings, Hall's theorem, Tutte's theorem, Konig's theorem, Petersen's theorem, algorithms for matching and weighted matching (in both bipartite and general graphs), factors of graphs (decompositions of the complete graph), Tutte's f- factor theorem, Extremal problems- Independent sets and covering numbers							
UNIT-IV (10 Hrs)	Colorings- Brooks theorem, the greedy algorithm, the Welsh-Powell bound, critical graphs, chromatic polynomials, girth and chromatic number, Vizing's theorem, Graphs on surfaces- Planar graphs, duality, Euler's formula, Kuratowski's theorem, toroidal graphs, 2-cell embeddings, graphs on other surfaces.							
UNIT-V (10 Hrs)	Directed graphs- Tournaments, directed paths and cycles, connectivity and strongly connected digraphs, Networks and flows- Flow cuts, max flow min cut theorem, Selected topics- Dominating sets, the reconstruction problem.							

Text Books:	
1.	Introduction to Graph Theory, Douglas B. West, Prentice Hall of India
2.	Graph Theory with Applications to Engineering and Computer Science, Narsingh Deo, Prentice-Hall
Reference Books:	
1.	Graph Theory, Frank Harary, Narosa
2.	Network Flows: Theory, Algorithms, and Applications, R.Ahuja, T. Magnanti, and J. Orlin, Prentice-Hall

Code	Category	L	T	P	C	I.M	E.M	Exam
M19RD1101	RD	2	0	0	2	25	75	3 Hrs.
RESEARCH METHODOLOGY AND IPR								
(Common to CST,CS,PSA, IT & CAD)								
Course Objectives:								
1.	To bring awareness on Research Methodology and research ethics.							
2.	Familiarize the concepts of IPR.							
Course Outcomes								
S.No	Outcome							Knowledge Level
1.	Analyze research related information							K4
2.	Formulate a Research Proposals and Publish papers with research ethics							K6
3.	Award for Intellectual Property Rights like Patents, Trade and Copyrights							K5
4.	Analyze Various Intellectual Property Rights							K4
5.	AssessNew Developments of IPRs in National and International level							K5
SYLLABUS								
UNIT-I (6 Hrs)	Meaning of research problem, Sources of research problem, Criteria Characteristics of a good research problem, Errors in selecting a research problem, Scope and objectives of research problem. Approaches of investigation of solutions for research problem, data collection, analysis, interpretation, Necessary instrumentations							
UNIT-II (6 Hrs)	Effective literature studies approaches, analysis Plagiarism, Research ethics, Effective technical writing, how to write report, Paper Developing a Research Proposal, Format of research proposal, a presentation and assessment by a review committee							
UNIT-III (6 Hrs)	Nature of Intellectual Property: Patents, Designs, Trade and Copyright. Process of Patenting and Development: technological research, innovation, patenting, development. International Scenario: International cooperation on Intellectual Property. Procedure for grants of patents, Patenting under PCT.							
UNIT-IV (4 Hrs)	Patent Rights: Scope of Patent Rights. Licensing and transfer of technology. Patent information and databases. Geographical Indications.							
UNIT-V (6 Hrs)	New Developments in IPR: Administration of Patent System. New developments in IPR; IPR of Biological Systems, Computer Software etc. Traditional knowledge Case Studies, IPR and IITs.							
Text Books:								
1.	Stuart Melville and Wayne Goddard, —Research methodology: an introduction for science & engineering students’l							
2.	Wayne Goddard and Stuart Melville, —Research Methodology: An Introductionll							
3.	Ranjit Kumar, 2nd Edition, —Research Methodology: A Step by Step Guide for beginnersl							

Reference Books:

1.	Halbert, —Resisting Intellectual Property, Taylor & Francis Ltd, 2007.
2.	Mayall, —Industrial Design, McGraw Hill, 1992.
3.	Niebel, —Product Design, McGraw Hill, 1974.
4.	Asimov, —Introduction to Design, Prentice Hall, 1962.
5.	Robert P. Merges, Peter S. Menell, Mark A. Lemley, — Intellectual Property in New Technological Age, 2016.
6.	T. Ramappa, —Intellectual Property Rights Under WTO, S. Chand, 2008

Code	Category	L	T	P	C	I.M	E.M	Exam
MTIT1113	PC	0	0	4	2	25	75	3 Hrs.
ADVANCED DATA STRUCTURES LAB								
Course Objectives:								
1.	The course is designed to develop skills to design and analyze different Algorithms							
2.	It enables them to gain knowledge in practical applications of data structures .							
Course Outcomes								
S.No	Outcome							Knowledge Level
1.	Be capable to Develop solutions for a range of problems using object oriented programming.							K6
2.	Be capable to Identity the appropriate data structure for given problem							K3
3.	Be capable to Analyze performance of algorithms.							K4
LIST OF EXPERIMENTS								
1.	Perform various operations on AVL Trees							
2.	Perform various operations on BST							
3.	Implementation of Static Hashing							
4.	Implementation of Huffman coding							
5.	Implementation of B Tree.							
6.	Consider telephone book database of N clients. Make use of a hash table implementation to quickly look up client,,s telephone number.							
7	Implement all the functions of a dictionary (ADT) using hashing. Data: Set of (key, value) pairs, Keys are mapped to values, Keys must be comparable, Keys must be unique Standard Operations: Insert(key, value), Find(key), Delete(key)							
8	For given set of elements create skip list. Find the element in the set that is closest to some given value.							
9	Implement KMP algorithm for Pattern Matching							
10	Implement Boyer-Moore algorithm for Pattern Matching							
11	Implement Naïve string matching algorithm.							
12	Implement insertion, deletion, display and search operation in m-way B tree (i.e. a nonleaf node can have atmost m children) for the given data as integers (Test the program for m=3, 5, 7).							
13	Implementation of Skiplists							
Reference Books:								
1.	Algorithms, Data Structures, and Problem Solving with C++ by Mark Allen Weiss							
2.	Data Structures and Algorithms in C++ by Brijendra Kumar Joshi							

Code	Category	L	T	P	C	I.M	E.M	Exam
M19IT1114	PC	0	0	4	2	25	75	3 Hrs.
COMPUTING LAB								
Course Objectives:								
1.	To introduce Python programming language through its core language basics and program design techniques suitable for modern applications.							
2.	To understand the wide range of programming facilities available in Python covering graphics, GUI, data visualization and Databases.							
3.	To utilize high-performance programming constructs available in Python to develop solutions in real life scenarios using AI.							
Course Outcomes								
S.No	Outcome							Knowledge Level
1.	Design real life situational problems and think creatively about solutions of them.							K6
2.	Formulate a solution clearly and accurately in a program using Python.							K6
3.	Inspect the best features of Python to program real life problems using AI.							K4
Artificial Intelligence Specialization :								
1.	Write a python program to print the multiplication table forthe given number.?							
2.	Write a python program to check whether the given number is prime or not.							
3.	Write a python program to find factorial of the given number.							
4.	Write a python program to implement simple Chatbot.							
5.	Write a python program to implement List operations (Nested List, Length, Concatenation, Membership, Iteration, Indexing and Slicing).							
6.	Write a python program to implement List methods (Add, Append, Extend & Delete).							
7.	Write a python program to Illustrate Different Set Operations.							
8.	Write a python program to generate Calendar for the given month andyear.							
9.	Write a python program to implement Simple Calculator program.							
10.	Write a python program to Add TwoMatrices.							
11.	Write a python program to Transpose a Matrix.							
12.	Write a python program to remove punctuations from the givenstring?							
13.	Implementation Write a python program to sort the sentence in alphabetical order.							
14.	Write a python program to implement Breadth First Search Traversal.							
15.	Write a python program to implement Water Jug Problem.							
16.	Write a program to implement Hangman game using python.							
17.	Write a program to implement Tic-Tac-Toe game using python.							
18.	Write a python program to remove stop words for a given passage from a text file using NLTK.							
19.	Write a python program to implement stemming for a given sentence using NLTK.							
20.	Write a python program to POS (Parts of Speech) tagging for the give sentence using NLTK?							
Reference Books:								
1. Let us Python, YashavantKanetkar BPB publications								
2. Artificial Intelligence with Python: A Comprehensive Guide to Building Intelligent Apps for Python Beginners and Developers, Prateek Joshi Packt Publishing								

Big Data Specialization :

S.No	Details
1	(i)Perform setting up and Installing Hadoop in its three operating modes: Standalone, Pseudo distributed, Fully distributed (ii)Use web based tools to monitor your Hadoop setup.
2	Implement the following file management tasks in Hadoop: • Adding files and directories • Retrieving files • Deleting files
3	Run a basic Word Count Map Reduce program to understand Map Reduce Paradigm.
4	Write a Map Reduce program that mines weather data. Weather sensors collecting data every hour at many locations across the globe gather a large volume of log data, which is a good candidate for analysis with MapReduce, since it is semi structured and record-oriented.
5	Implement Matrix Multiplication with Hadoop Map Reduce
6	Install and Run Pig then write Pig Latin scripts to sort, group, join, project, and filter Your data.
7	Install and Run Hive then use Hive to create, alter, and drop databases, tables, views, functions, and indexes

Reference Books:

1. Hadoop Map Reduce Cookbook, Srinath Perera&ThilinaGunarathne, 2013, PACKT PUBLISHING

Cryptography Specialization :

Exercise	Exercise Details
1	Write a Java program to perform encryption and decryption using the following algorithms: a) Ceaser Cipher b) Substitution Cipher c) Hill Cipher
2	Write a Java program to implement the 3 DES and AES algorithms.
3	Write a JAVA program to implement the BlowFish algorithm
4	Implement MD-5 using Java
5	Write a Java program to implement RSA (2048 Key Length) Algorithm.
6	Implement the Diffie-Hellman Key Exchange mechanism using HTML and JavaScript. Consider the end user as one of the parties (Alice) and the JavaScript application as other party (bob).
7	Calculate the message digest of a text using the SHA-2 algorithm in JAVA.

Reference Books:

1. Beginning Cryptography with Java, David Hook Wiley India Private Limited

Code	Category	L	T	P	C	I.M	E.M	Exam
M19AC0001	AC	2	0	0	0	0	0	--
ENGLISH FOR RESEARCH PAPER WRITING								
Course Objectives:								
1.	Understand how to improve your writing skills and level of readability							
2.	Learn about what to write in each section.							
3.	Understand the skills needed when writing a Title.							
4.	Ensure the good quality of paper at very first-time submission							
Course Outcomes								
S.No	Outcome							Knowledge Level
1.	Understand that how to improve your writing skills and level of readability							K2
2.	Learn about what to write in each section							K2
3.	Understand the skills needed when writing a Title Ensure the good quality of paper at very first time submission							K2
SYLLABUS								
UNIT-I (4Hrs)	Planning and Preparation, Word Order, Breaking up long sentences, Structuring Paragraphs and Sentences, Being Concise and Removing Redundancy, Avoiding Ambiguity and Vagueness							
UNIT-II (4Hrs)	Clarifying Who Did What, Highlighting Your Findings, Hedging And Criticizing, Paraphrasing and Plagiarism, Sections of a Paper.							
UNIT-III (4Hrs)	Abstracts, Introduction, Review of the Literature, Methods, Results, Discussion, Conclusions, The Final Check.							
UNIT-IV (4Hrs)	Key skills are needed when writing a Title, key skills are needed when writing an abstract, key skills are needed when writing an introduction, skills needed when writing a Review of the Literature, skills are needed when writing the Methods.							
UNIT-V (4Hrs)	skills needed when writing the Results, skills are needed when writing the Discussion, skills are needed when writing the Conclusions, useful phrases, how to ensure paper is as good as it could possibly be the first- time submission							
Text Books:								
1.	Goldbort R (2006) Writing for Science, Yale University Press (available on Google Books)							
2.	Day R (2006) How to Write and Publish a Scientific Paper, Cambridge University Press							
Reference Books:								
1.	Highman N (1998), Handbook of Writing for the Mathematical Sciences, SIAM. Highman’s book.							
2.	Adrian Wallwork , English for Writing Research Papers, Springer New York Dordrecht Heidelberg London, 2011							

Code	Category	L	T	P	C	I.M	E.M	Exam
M19AC0002	AC	2	0	0	0	0	0	--
DISASTER MANAGEMENT								
Course Objectives:								
1.	Learn to demonstrate a critical understanding of key concepts in disaster risk reduction and humanitarian response.							
2.	Critically evaluate disaster risk reduction and humanitarian response policy and practice from multiple perspectives.							
3.	Develop an understanding of standards of humanitarian response and practical relevance in specific types of disasters and conflict situations.							
4.	Critically understand the strengths and weaknesses of disaster management approaches, planning & programming in different countries, particularly their home country or the countries they work in.							
SYLLABUS								
UNIT-I (4Hrs)	Disaster: Definition, Factors And Significance; Difference Between Hazard And Disaster; Natural And Manmade Disasters: Difference, Nature, Types And Magnitude.							
UNIT-II (4Hrs)	Economic Damage, Loss Of Human And Animal Life, Destruction Of Ecosystem. Natural Disasters: Earthquakes, Volcanisms, Cyclones, Tsunamis, Floods, Droughts And Famines, Landslides And Avalanches, Man-made disaster: Nuclear Reactor Meltdown, Industrial Accidents, Oil Slicks And Spills, Outbreaks Of Disease And Epidemics, War And Conflicts							
UNIT-III (4Hrs)	Disaster Prone Areas In India: Study Of Seismic Zones; Areas Prone To Floods And Droughts, Landslides And Avalanches; Areas Prone To Cyclonic And Coastal Hazards With Special Reference To Tsunami; Post-Disaster Diseases And Epidemics.							
UNIT-IV (4Hrs)	Disaster Preparedness And Management Preparedness: Monitoring Of Phenomena Triggering A Disaster Or Hazard; Evaluation Of Risk: Application Of Remote Sensing, Data From Meteorological And Other Agencies, Media Reports: Governmental And Community Preparedness.							
UNIT-V (4Hrs)	Risk Assessment Disaster Risk: Concept And Elements, Disaster Risk Reduction, Global And National Disaster Risk Situation. Techniques Of Risk Assessment, Global Co-Operation In Risk Assessment And Warning, People's Participation In Risk Assessment. Strategies for Survival. Disaster Mitigation Meaning, Concept And Strategies Of Disaster Mitigation, Emerging Trends In Mitigation. Structural Mitigation And Non-Structural Mitigation, Programs Of Disaster Mitigation In India.							
Text Books:								
1.	R. Nishith, Singh AK, Disaster Management in India: Perspectives, issues and strategies New Royal Book Company.							

2.	Sahni, PardeepEt.Al. (Eds.),Disaster Mitigation Experiences And Reflections, Prentice Hall Of India, New Delhi.
3.	Goel S. L. Disaster Administration And Management Text And Case Studies ,Deep &Deep Publication Pvt. Ltd., New Delhi.

Code	Category	L	T	P	C	I.M	E.M	Exam
M19AC0003	AC	2	0	0	0	0	0	--
SANSKRIT FOR TECHNICAL KNOWLEDGE								
Course Objectives:								
1.	To get a working knowledge in illustrious Sanskrit, the scientific language in the world							
2.	Learning of Sanskrit to improve brain functioning							
3.	Learning of Sanskrit to develop the logic in mathematics, science & other subjects enhancing the memory power							
4.	The engineering scholars equipped with Sanskrit will be able to explore the huge knowledge from ancient literature							
Course Outcomes								
S.No	Outcome							Knowledge Level
1	Understanding basic Sanskrit language.							K2
2	Ancient Sanskrit literature about science & technology can be understood.							K2
3	Being a logical language will help to develop logic in students.							K6
SYLLABUS								
UNIT-I (7Hrs)	Alphabets in Sanskrit, Past/Present/Future Tense, Simple Sentences							
UNIT-II (7Hrs)	Order, Introduction of roots, Technical information about Sanskrit Literature							
UNIT-III (7Hrs)	Technical concepts of Engineering-Electrical, Mechanical, Architecture, Mathematics							
Text Books:								
1.	“AbhyasaPustakam” – Dr.Vishwas, Samskrita-Bharati Publication, New Delhi							
2.	“Teach Yourself Sanskrit” PrathamaDiksha-VempatiKutumbasastry, Rashtriya Sanskrit Sansthanam, New Delhi Publication							
3.	“India’s Glorious Scientific Tradition” Suresh Soni, Ocean books (P) Ltd., New Delhi							

Code	Category	L	T	P	C	I.M	E.M	Exam
M19AC0004	AC	2	0	0	0	0	0	--
VALUE EDUCATION								
Course Objectives:								
1.	Understand the value of education and self- development							
2.	Imbibe good values in students							
3.	Let should know about the importance of character							
Course Outcomes								
S.No	Outcome							Knowledge Level
1	Knowledge of self-development							K1
2	Learn the importance of Human values							K2
3	Developing the overall personality							K3
SYLLABUS								
UNIT-I (4Hrs)	Values and self-development –Social values and individual attitudes. Work ethics, Indian vision of humanism.Moral and non- moral valuation. Standards and principles. Value judgements.							
UNIT-II (4Hrs)	Importance of cultivation of values. Sense of duty. Devotion, Self-reliance. Confidence, Concentration. Truthfulness, Cleanliness. Honesty, Humanity. Power of faith, National Unity, Patriotism, Love for nature ,Discipline							
UNIT-III (4Hrs)	Personality and Behavior Development - Soul and Scientific attitude. Positive Thinking. Integrity and discipline. Punctuality, Love and Kindness. Avoid fault Thinking. Free from anger, Dignity of labour. Universal brotherhood and religious tolerance. True friendship. Happiness Vs suffering, love for truth. Aware of self-destructive habits. Association and Cooperation, Doing best for saving nature							
UNIT-IV (4Hrs)	Character and Competence –Holy books vs Blind faith.Self-management and Good health. Science of reincarnation. Equality, Nonviolence ,Humility, Role of Women. All religions and same message. Mind your Mind, Self-control. Honesty, Studying effectively.							
Text Books:								
1.	Chakroborty, S.K. “Values and Ethics for organizations Theory and practice”, Oxford University Press, New Delhi							

Code	Category	L	T	P	C	I.M	E.M	Exam
M19AC0005	AC	2	0	0	0	0	0	--
CONSTITUTION OF INDIA								
Course Objectives:								
1.	Understand the premises informing the twin themes of liberty and freedom from a civil rights perspective.							
2.	To address the growth of Indian opinion regarding modern Indian intellectuals’ constitutional role and entitlement to civil and economic rights as well as the emergence of nationhood in the early years of Indian nationalism							
3.	To address the role of socialism in India after the commencement of the Bolshevik Revolution in 1917 and its impact on the initial drafting of the Indian Constitution							
Course Outcomes								
S.No	Outcome							Knowledge Level
1	Discuss the growth of the demand for civil rights in India for the bulk of Indians before the arrival of Gandhi in Indian politics.							K2
2	Discuss the intellectual origins of the framework of argument that informed the conceptualization of social reforms leading to revolution in India.							K2
3	Discuss the circumstances surrounding the foundation of the Congress Socialist Party [CSP] under the leadership of Jawaharlal Nehru and the eventual failure of the proposal of direct elections through adult suffrage in the Indian Constitution.							K2
4	Discuss the passage of the Hindu Code Bill of 1956.							K2
SYLLABUS								
UNIT-I (4Hrs)	History of Making of the Indian Constitution: History , Drafting Committee, (Composition & Working)							
UNIT-II (4Hrs)	Philosophy of the Indian Constitution: Preamble ,Salient Features							
UNIT-III (4Hrs)	Fundamental Rights,Right to Equality, Right to Freedom,Right against Exploitation, Right to Freedom of Religion,Cultural and Educational Rights,Right to Constitutional Remedies, Directive Principles of State Policy, Fundamental Duties							
UNIT-IV (4Hrs)	Organs of Governance: Parliament, Composition, Qualifications and Disqualifications ,Powers and Functions, Executive, President , Governor, Council of Ministers, Judiciary, Appointment and Transfer of Judges, Qualifications, Powers and Functions Local Administration: District’s Administration head: Role and Importance, Municipalities: Introduction, Mayor and role of Elected Representative, CEO of Municipal Corporation, Pachayati raj: Introduction, PRI: Zilla Panchayat. Elected officials and their roles, CEO Zilla Panchayat:							

	Position and role. Block level: Organizational Hierarchy (Different departments), Village level: Role of Elected and Appointed officials, Importance of grass root democracy
UNIT-V (4Hrs)	Election Commission: Election Commission: Role and Functioning. Chief Election Commissioner and Election Commissioners. State Election Commission: Role and Functioning. Institute and Bodies for the welfare of SC/ST/OBC and women.
Text Books:	
1.	The Constitution of India, 1950 (Bare Act), Government Publication.
2.	Dr. S. N. Busi, Dr. B. R. Ambedkar framing of Indian Constitution, 1st Edition, 2015.
3.	M. P. Jain, Indian Constitution Law, 7th Edn., LexisNexis, 2014.
4.	D.D. Basu, Introduction to the Constitution of India, LexisNexis, 2015.

Code	Category	L	T	P	C	I.M	E.M	Exam
M19AC0006	AC	2	0	0	0	0	0	--
PEDAGOGY STUDIES								
SYLLABUS								
UNIT-I (4Hrs)	Introduction and Methodology: Aims and rationale, Policy background, Conceptual framework and terminology Theories of learning, Curriculum, Teacher education. Conceptual framework, Research questions. Overview of methodology and Searching.							
UNIT-II (4Hrs)	Thematic overview: Pedagogical practices are being used by teachers in formal and informal classrooms in developing countries, Curriculum, Teacher education.							
UNIT-III (4Hrs)	Evidence on the effectiveness of pedagogical practices, Methodology for the in depth stage: quality assessment of included studies. How can teacher education (curriculum and practicum) and the school curriculum and guidance materials best support effective pedagogy.							
UNIT-IV (4Hrs)	Theory of change, Strength and nature of the body of evidence for effective pedagogical practices Pedagogic theory and pedagogical approaches. Teachers’ attitudes and beliefs and Pedagogic strategies. Professional development: alignment with classroom practices and follow-up support							
UNIT-V (4Hrs)	Research gaps and future directions: Research design, Contexts, Pedagogy, Teacher education, Curriculum and assessment, Dissemination and research impact.							
Text Books:								
1.	Ackers J, Hardman F (2001) Classroom interaction in Kenyan primary schools, Compare, 31 (2): 245-261.							
2.	Agrawal M (2004) Curricular reform in schools: The importance of evaluation, Journal of Curriculum Studies, 36 (3): 361-379.							
3.	Akyeampong K (2003) Teacher training in Ghana - does it count? Multi-site teacher education research project (MUSTER) country report 1. London: DFID.							
4.	Akyeampong K, Lussier K, Pryor J, Westbrook J (2013) Improving teaching and learning of basic maths and reading in Africa: Does teacher preparation count? International Journal of Educational Development, 33 (3): 272–282.							
5.	Alexander RJ (2001) Culture and pedagogy: International comparisons in primary education. Oxford and Boston: Blackwell.							
6.	Chavan M (2003) Read India: A mass scale, rapid, „learning to read“ campaign.							
7.	www.pratham.org/images/resource%20working%20paper%202.pdf.							

Code	Category	L	T	P	C	I.M	E.M	Exam
M19AC0007	AC	2	0	0	0	0	0	--
STRESS MANAGEMENT BY YOGA								
Course Objectives:								
1.	To achieve overall health of body and mind							
2.	To overcome stress							
Course Outcomes								
S.No	Outcome							Knowledge Level
1	Develop a healthy mind in a healthy body thus improving social health also.							K2
2	Improve efficiency							K2
SYLLABUS								
UNIT-I (7Hrs)	Definitions of Eight parts of yoga (Ashtanga)							
UNIT-II (7Hrs)	Yam and Niyam. Do's and Don'ts in life. i) Ahinsa, satya, astheya, brahmacharya and aparigraha ii) Shaucha, santosh, tapa, swadhyay, ishvarapranidhana							
UNIT-III (7Hrs)	Asan and Pranayam i) Various yog poses and their benefits for mind & body ii)Regularization of breathing techniques and its effects-Types pranayama							
Text Books:								
1.	‘Yogic Asanas for Group Training-Part-I’ : Janardan Swami YogabhyasiMandal, Nagpur							
2.	“Raja Yoga or conquering the Internal Nature” by Swami Vivekananda, AdvaitaAshrama (Publication Department), Kolkata							

Code	Category	L	T	P	C	I.M	E.M	Exam
M19AC0008	AC	2	0	0	0	0	0	--
PERSONALITY DEVELOPMENT THROUGH LIFEENLIGHTENMENT SKILLS								
Course Objectives:								
1.	To learn to achieve the highest goal happily							
2.	To become a person with a stable mind, pleasing personality and determination							
3.	To awaken wisdom in students							
Course Outcomes								
S.No	Outcome							Knowledge Level
1	Study of Shrimad-Bhagwad-Geeta will help the student in developing his personality and achieve the highest goal in life.							K2
2	The person who has studied Geeta will lead the nation and mankind to peace and prosperity.							K2
3	Study of Neetishatakam will help in developing versatile personality of students.							K2
SYLLABUS								
UNIT-I (7Hrs)	Neetisatakam-Holistic development of personality Verses- 19,20,21,22 (wisdom), Verses- 29,31,32 (pride & heroism) Verses- 26,28,63,65 (virtue),Verses- 52,53,59 (don'ts),Verses- 71,73,75,78 (do's)							
UNIT-II (7Hrs)	Approach to day to day work and duties. ShrimadBhagwadGeeta : Chapter 2-Verses: 41, 47,48 Chapter 3-Verses 13, 21, 27, 35, Chapter 6-Verses 5,13,17, 23, 35, Chapter 18-Verses 45, 46, 48.							
UNIT-III (7Hrs)	Statements of basic knowledge, ShrimadBhagwadGeeta: Chapter2-Verses 56, 62, 68, Chapter 12 -Verses 13, 14, 15, 16,17, 18, Personality of Role model. ShrimadBhagwadGeeta: Chapter2-Verses 17, Chapter 3-Verses 36,37,42, Chapter 4-Verses 18, 38,39, Chapter18 – Verses 37,38,63							
Text Books:								
1.	‘Srimad Bhagavad Gita” by Swami SwarupanandaAdvaita Ashram (Publication Department), Kolkata							
2.	Bhartrihari’s Three Satakam (Niti-sringar-vairagya) by P.Gopinath,Rashtriya Sanskrit Sansthanam, New Delhi.							



SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (AUTONOMOUS)

(Affiliated to JNTUK, Kakinada), (Recognized by AICTE, New Delhi)

Accredited by NAAC with 'A' Grade

CHINNA AMIRAM (P.O):: BHIMAVARAM :: W.G.Dt., A.P., INDIA :: PIN: 534 204

SCHEME OF INSTRUCTION & EXAMINATION

(Regulation R19)

M.TECH (INFORMATION TECHNOLOGY)

DEPARTMENT OF INFORMATION TECHNOLOGY

(With effect from 2019-2020 Admitted Batch onwards)

II-SEMESTER

Subject Code.	Name of the Subject	Category.	C	L	T	P	Int. Marks	External Marks	Total Marks
M19IT1201	Advanced Algorithms	PC	3	3	0	0	25	75	100
M19IT1202	Full Stack Technologies	PC	3	3	0	0	25	75	100
#PE-III	Program Elective-III	PE	3	3	0	0	25	75	100
#PE-IV	Program Elective-IV	PE	3	3	0	0	25	75	100
M19IT1213	Advance Algorithms Lab	PC	2	0	0	4	25	75	100
M19IT1214	Full Stack Technologies Lab	PC	2	0	0	4	25	75	100
M19IT1215	Mini Project with Seminar	MP	2	0	0	4	100	--	100
#AC-2	Audit Course-2*	AC	0	2	0	0	0	0	0
Total			18	14	0	12	250	450	700

	Code	Course
#PE-III	M19IT1203	Machine Learning
	M19IT1204	DevOps
	M19IT1205	Advanced Network Principles and Protocols
	M19IT1206	Distributed Computing
	M19IT1207	Social Network Analytics
#PE-IV	M19IT1208	Digital Image Processing
	M19IT1209	Block Chain Technologies
	M19IT1210	Data Science
	M19IT1211	Soft Computing
	M19IT1212	Natural Language Processing

Code	Category	L	T	P	C	I.M	E.M	Exam
M19IT1201	PC	3	0	0	3	25	75	3 Hrs.
ADVANCED ALGORITHMS								
Course Objectives:								
1.	To Understand The fundamental design, analysis, and implementation of basic data structures.							
2.	To Understand Basic concepts in the specification and analysis of programs.							
3.	To Understand Principles for good program design, especially the uses of data abstraction.							
4.	To Understand Significance of algorithms in the computer field							
5.	To Understand Various aspects of algorithm development							
6.	To Understand Qualities of a good solution							
Course Outcomes After completion of course, students would be able to:								
S.No	Outcome							Knowledge Level
1.	Analyze advanced algorithms and solve practical problem using them.							K4
2.	Categorize the different problems in various classes according to their complexity.							K4
3.	Design and build solutions for a real world problem by applying relevant distributions							K5
SYLLABUS								
UNIT-I (10Hrs)	Sorting: Review of various sorting algorithms, topological sorting Graph: Definitions and Elementary Algorithms: Shortest path by BFS, shortest path in edge-weighted case (Dijkasra's), depth-first search and computation of strongly connected components, emphasis on correctness proof of the algorithm and time/space analysis.							
UNIT-II (10Hrs)	Matroids: Introduction to greedy paradigm, algorithm to compute a maximum weight maximal independent set. Application to MST. Greedy Algorithms: Theoretical foundations for greedy methods, A Task scheduling problem.							
UNIT-III (10Hrs)	Flow-Networks: Max flow-mincut theorem, Ford-Fulkerson Method to compute maximum flow, Edmond-Karp maximum-flow algorithm. Matrix Computations: Strassen's algorithm and introduction to divide and conquer paradigm, inverse of a triangular matrix.							
UNIT-IV (10Hrs)	Shortest Path in Graphs: Floyd-Warshall algorithm and introduction to dynamic programming paradigm. More examples of dynamic programming. Modulo Representation of integers/polynomials: Chinese Remainder Theorem, Conversion between base-representation and modulo-representation. Application: Interpolation problem. Polynomials and the FFT: Representation of of polynomials, The DFT and FFT, Efficient FFT Implementation							
UNIT-V (10Hrs)	Linear Programming: Geometry of the feasibility region and Simplex algorithm Completeness: Examples, proof of NP-hardness and NP-completeness. One or more of the following topics based on time and interest Approximation algorithms, Randomized Algorithms, Interior Point Method.							
Text Books:								
1.	Cormen, Leiserson, Rivest, Stein, "Introduction to Algorithms", Tata Mcgraw Hill Publishers, 2001.							

2.	Algorithm Design: Foundations, Analysis and Internet examples, M.T. Goodrich and R.Tomassia, John Wiley and sons.
Reference Books:	
1.	Aho, Hopcroft, Ullman, "The Design and Analysis of Computer Algorithms", Pearson Education, 1974. 2. Kleinberg and Tardos, "Algorithm Design", Pearson Education, 2006.

Code	Category	L	T	P	C	I.M	E.M	Exam
M19IT1202	PC	3	0	0	3	25	75	3 Hrs.
FULL STACK TECHNOLOGIES								
Course Objectives:								
1.	Translate user requirements into the overall architecture and implementation of new systems and Manage Project and coordinate with the Client.							
2.	Write backend code in Python/Java, PHP languages and Writing optimized front end code HTML and JavaScript.							
3.	Understand, create and debug database related queries and Create test code to validate the applications against client requirement.							
4.	Monitor the performance of web applications & infrastructure and Troubleshooting web application with a fast and accurate a resolution.							
Course Outcomes After completion of course, students would be able to:								
S.No	Outcome							Knowledge Level
1.	Identify the Basic Concepts of Web & Markup Languages							K3
2.	Develop web Applications using Scripting Languages & Frameworks							K3
3.	Creating & Running Applications using JSP libraries							K4
4.	Creating Our First Controller Working with and Displaying in Angular Js and Nested Forms with ng-form							K3
5.	Creating & Running Back-end scripts & Connecting to Databases							K4
SYLLABUS								
UNIT-I (10Hrs)	HTML : Web Essentials: Clients, Servers, and Communication. The Internet-Basic Internet Protocols -The World Wide Web-HTTP request message-response message-Web Clients Web Servers. Markup Languages: XHTML an Introduction to HTML, History, Versions, Basic, XHTML Syntax and Semantics Some Fundamental HTML Elements-Relative URLs-Lists-tables-Frames-Forms-HTML 5.0.							
UNIT-II (10Hrs)	Cascading Style Sheets (CSS) : Style Sheets: CSS-Introduction to Cascading Style Sheets-Features-Core Syntax-Style Sheets and HTML- Style Rule Cascading and Inheritance-Text Properties-Box Model Normal Flow Box Layout beyond the Normal Flow-CSS3.0, Boot strap basics, Boot strap CSS3, Introduction to Java Script, Jscript basics, JScripts objects, JSON, Don.							
UNIT-III (10Hrs)	Jscript : Separating Programming and Presentation: JSP Technology, Introduction to JSP and Servlets-Running JSP Applications, Basic JSP-JavaBeans Classes and JSP-Tag Libraries and Files-Support for the Model-View-Controller Paradigm- Mongo DB, JQuery, Mean stack Fundamentals.							
UNIT-IV (10Hrs)	Angular Js : Introducing AngularJS, Starting Out with AngularJS, Basic AngularJS, Directives and Controllers, AngularJS Modules, Creating First Controller, working with and Displaying, Arrays, more Directives, working with ng-repeat, Unit Testing in AngularJS, Forms, Inputs, and Services, Working with ng-model, Working with Forms, Leverage Data-Binding and Models, Form Validation and States, Error Handling with Forms, ngModelOptions, Nested Forms with ng-form, Other Form Controls.							

UNIT-V (10Hrs)	PHP Programming: Back – end- Scripts PHP, Working with PHP- Using variables, Using constants, Data types, Operators. Conditional & Control statements, Arrays, functions. Working with forms and Databases such as MySQL.
Text Books:	
1.	Jeffrey C. Jackson, "Web Technologies--A Computer Science Perspective", Pearson Education, 2006
2.	Robert. W. Sebesta, "Programming the World Wide Web", Fourth Edition, Pearson Education, 2007
3.	AngularJS: Up and Running Enhanced Productivity with Structured Web Apps By Brad Green, Shyam Seshadri Publisher: O'Reilly Media
Reference Books:	
1.	Learning React Functional Web Development with React and Redux By Alex Banks, Eve Porcello Publisher: O'Reilly Media
2.	Head First Java, 2nd Edition by Bert Bates, Kathy Sierra Publisher: O'Reilly Media, Inc

Code	Category	L	T	P	C	I.M	E.M	Exam
M19IT1203	PE	3	0	0	3	25	75	3 Hrs.
MACHINE LEARNING								
(Program Elective-III)								
Course Objectives:								
1.	Develop an appreciation for what is involved in learning from data.							
2.	Demonstrate a wide variety of learning algorithms.							
3.	Demonstrate how to apply a variety of learning algorithms to data.							
4.	Demonstrate how to perform evaluation of learning algorithms and model selection.							
Course Outcomes At the end of the course, student will be able to								
S.No	Outcome							Knowledge Level
1.	Domain Knowledge for Productive use of Machine Learning and Diversity of Data.							K4
2.	Demonstrate on Supervised and Computational Learning							K4
3.	Analyze on Statistics in learning techniques and Logistic Regression							K4
4.	Illustrate on Support Vector Machines and Perceptron Algorithm							K4
5.	Design a Multilayer Perceptron Networks and classification of decision tree							K3
SYLLABUS								
UNIT-I (10 Hrs)	Introduction-Towards Intelligent Machines, Well posed Problems, Example of Applications in diverse fields, Data Representation, Domain Knowledge for Productive use of Machine Learning, Diversity of Data: Structured / Unstructured, Forms of Learning, Machine Learning and Data Mining, Basic Linear Algebra in Machine Learning Techniques.							
UNIT-II (10 Hrs)	Supervised Learning- Rationale and Basics: Learning from Observations, Bias and Why Learning Works: Computational Learning Theory, Occam's Razor Principle and Over fitting Avoidance Heuristic Search in inductive Learning, Estimating Generalization Errors, Metrics for assessing regression, Metris for assessing classification.							
UNIT-III (10 Hrs)	Statistical Learning- Machine Learning and Inferential Statistical Analysis, Descriptive Statistics in learning techniques, Bayesian Reasoning: A probabilistic approach to inference, K-Nearest Neighbor Classifier. Discriminant functions and regression functions, Linear Regression with Least Square Error Criterion, Logistic Regression for Classification Tasks, Fisher's Linear Discriminant and Thresholding for Classification, Minimum Description Length Principle.							
UNIT-IV (10 Hrs)	Support Vector Machines (SVM)- Introduction, Linear Discriminant Functions for Binary Classification, Perceptron Algorithm, Large Margin Classifier for linearly seperable data, Linear Soft Margin Classifier for Overlapping Classes, Kernel Induced Feature Spaces, Nonlinear Classifier, Regression by Support vector Machines. Learning with Neural Networks: Towards Cognitive Machine, Neuron Models, Network Architectures, Perceptrons, Linear neuron and the Widrow-Hoff Learning Rule, The error correction delta rule.							

UNIT-V (10 Hrs)	Multilayer Perceptron Networks and error back propagation algorithm, Radial Basis Functions Networks. Decision Tree Learning: Introduction, Example of classification decision tree, measures of impurity for evaluating splits in decision trees, ID3, C4.5, and CART decision trees, pruning the tree, strengths and weakness of decision tree approach.
Text Books:	
1	Applied Machine Learning, M. Gopal, McGraw Hill Education
2	Machine Learning: A Probabilistic Perspective, Kevin Murphy, MIT Press 2012
3	The Elements of Statistical Learning, Trevor Hastie, Robert Tibshirani, Jerome Friedman, Springer 2009 (freely available online)
Reference Books:	
1	Pattern Recognition and Machine Learning, Christopher Bishop, Springer, 2007
2	Programming Collective Intelligence: Building Smart Web 2.0 Applications - Toby Segaran
3	Building Machine Learning Systems with Python - Willi Richert, Luis Pedro Coelho

Code	Category	L	T	P	C	I.M	E.M	Exam
M19IT1204	PE	3	0	0	3	25	75	3 Hrs.
DEVOPS								
(Program Elective-III)								
Course Objectives:								
1	DevOps improves collaboration and productivity by automating infrastructure and workflows and continuously measuring applications performance.							
Course Outcomes At the end of the course, student will be able to								
S.No	Outcome							Knowledge Level
1	Apply the principles of continuous development and deployment, automation of configuration management, inter-team collaboration, and IT service agility							K3
2	Identity DevOps & DevSecOps methodologies and their key concepts							K3
3	List out the types of version control systems, continuous integration tools, continuous monitoring tools, and cloud models							K4
4	Set up complete private infrastructure using version control systems and CI/CD tools							K4
SYLLABUS								
UNIT-I (10 Hrs)	Phases of Software Development life cycle. Values and principles of agile software development.							
UNIT-II (10 Hrs)	Fundamentals of DevOps: Architecture, Deployments, Orchestration, Need, Instance of applications, DevOps delivery pipeline, DevOps eco system.							
UNIT-III (10 Hrs)	DevOps adoption in projects: Technology aspects, Agiling capabilities, Tool stack implementation, People aspect, processes							
UNIT-IV (10 Hrs)	CI/CD: Introduction to Continuous Integration, Continuous Delivery and Deployment, Benefits of CI/CD, Metrics to track CICD practices							
UNIT-V (10 Hrs)	Devops Maturity Model: Key factors of DevOps maturity model, stages of Devops maturity model, DevOps maturity Assessment							
Text Books:								
1.	The DevOPS Handbook: How to Create World-Class Agility, Reliability, and Security in Technology Organizations by Gene Kim , John Willis , Patrick Debois , Jez Humb,O'Reilly publications							
2.	What is Devops? Infrastructure as code By in Mike Loukides ,O'Reilly publications.							
3.	Continuous Delivery: Reliable Software Releases Through Build, Test, and Deployment Automation, by Jez Humble and David Farley							
4.	Achieving DevOps: A Novel About Delivering the Best of Agile, DevOps, and Microservices by Dave Harrison, Knox Lively							
Reference Books:								
1.	Building a DevOps Culture by Mandi Walls, O'Reilly publications							

2.	The DevOps 2.0 Toolkit: Automating the Continuous Deployment Pipeline With Containerized Microservices by Viktor Farcic
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Code	Category	L	T	P	C	I.M	E.M	Exam
M19IT1205	PC	3	0	0	3	25	75	3 Hrs.
ADVANCED NETWORK PRINCIPLES AND PROTOCOLS								
(Program Elective-III)								
Course Objectives:								
1.	Understand the architecture of the Internet protocols as a layered model.							
2.	To understand the fundamentals of data transmission, encoding and multiplexing							
3.	To understand how the various components of wide area networks and local area networks work together							
Course Outcomes: After completion of course, students will be able to:								
S. No	Outcome							Knowledge Level
1.	Identify the different layers of TCP/IP protocol stack							K3
2	Analyze the Concepts of Network media and topologies, Network security concepts and Network management							K4
3	Identify the working principle of different protocols at different layers							K3
SYLLABUS								
UNIT-I (10 Hrs)	Introduction to Networks - Application of Networks - Architecture Topology Switching - SLIP, PPP -ALOHA protocols, CSMA/CD, IEEE 802.3, 802.4, 802.5							
UNIT-II (10 Hrs)	Network Layer Issues- Routing, Congestion control- Internetworking - Issues, Address Learning Bridges, Spanning tree, Source routing, Bridges, Routers, Gateway.							
UNIT-III (10 Hrs)	Network Protocol- IP datagram - hop by hop routing, ARP, RARP, DHCP -Sub net Addressing, Address Masking, ICMP, RIP, RIPv2, OSPF, DNS, LAN and WAN Multicast.							
UNIT-IV (10 Hrs)	Transport Layer- Design issues, Connection Management, Transmission Control Protocol (TCP) - User Datagram Protocol (UDP).							
UNIT-V (10 Hrs)	Application Layer Protocol- Telnet - TFTP - FTP - SMTP - Ping Finger, Bootstrap Network Time Protocol- SNMP.							
Text Books:								
1.	Andrew S. Tanenbaum and David J. Wetherall, “Computer Networks”, 5th Edition, Pearson, 2011							
2.	William Stallings, "Data and Computer Communications”, 9th Edition, Pearson, 2011							
Reference Books:								
1.	W Richard Stevens and G. Gabrani, "TCP/IP Illustrated - Volume I, The protocols", Pearson Education, 2009							

Code	Category	L	T	P	C	I.M	E.M	Exam
M19IT1206	PE	3	0	0	3	25	75	3 Hrs.
DISTRIBUTED COMPUTING								
(Program Elective-III)								
Course Objectives:								
1	To understand the foundations of distributed systems.							
2	To learn issues related to clock Synchronization and the need for global state in distributed systems.							
3	To learn distributed mutual exclusion and deadlock detection algorithms.							
4	To understand the significance of agreement, fault tolerance and recovery protocols in Distributed Systems.							
5	To learn the characteristics of peer-to-peer and distributed shared memory systems.							
S.No	Outcome							Knowledge Level
1	Elucidate the foundations and issues of distributed systems							K3
2	Analyze various synchronization issues and global state for distributed systems.							K4
3	Analyze the Mutual Exclusion and Deadlock detection algorithms in distributed systems							K4
4	Analyze the agreement protocols and fault tolerance mechanisms in distributed systems.							K4
5	Analyze the features of peer-to-peer and distributed shared memory systems							K4
SYLLABUS								
UNIT-I (10 Hrs)	INTRODUCTION: Definition, Relation to computer system components, Motivation, Relation to parallel systems, Message-passing systems versus shared memory systems, Primitives for distributed communication, Synchronous versus asynchronous executions, Design issues and challenges. A model of distributed computations: A distributed program, A model of distributed executions, Models of communication networks, Global state, Cuts, Past and future cones of an event, Models of process communications. Logical Time: A framework for a system of logical clocks, Scalar time, Vector time, Physical clock synchronization: NTP.							
UNIT-II (10 Hrs)	MESSAGE ORDERING & SNAPSHOTS: Message ordering and group communication: Message ordering paradigms, Asynchronous execution with synchronous communication, Synchronous program order on an asynchronous system, Group communication, Causal order (CO), Total order. Global state and snapshot recording algorithms: Introduction, System model and definitions, Snapshot algorithms for FIFO channels							
UNIT-III (10 Hrs)	DISTRIBUTED MUTEX & DEADLOCK: Distributed mutual exclusion algorithms: Introduction, Preliminaries, Lamport’s algorithm, RicartAgrawala algorithm, Maekawa’s algorithm, Suzuki–Kasami’s broadcast algorithm. Deadlock detection in distributed systems: Introduction, System model, Preliminaries, Models of deadlocks, Knapp’s classification, Algorithms for the single resource model, the AND model and the OR model.							

UNIT-IV (10 Hrs)	RECOVERY & CONSENSUS: Check pointing and rollback recovery: Introduction, Background and definitions, Issues in failure recovery, Checkpoint-based recovery, Log-based rollback recovery, coordinated check pointing algorithm, Algorithm for asynchronous check pointing and recovery. Consensus and agreement algorithms: Problem definition, Overview of results, Agreement in a failure –free system, Agreement in synchronous systems with failures.
UNIT-V (10 Hrs)	P2P & DISTRIBUTED SHARED MEMORY: Peer-to-peer computing and overlay graphs: Introduction, Data indexing and overlays, Chord – Content addressable networks, Tapestry. Distributed shared memory: Abstraction and advantages, Memory consistency models, Shared memory Mutual Exclusion.
Text Books:	
1.	Distributed computing: principles, algorithms, and systems, Kshemkalyani, Ajay D., and Mukesh Singhal. Cambridge University Press, 2011.
2.	Distributed Systems Concepts and Design, George Coulouris, Jean Dollimore and Tim Kindberg , Fifth Edition, Pearson Education, 2012.
Reference Books:	
1.	“Distributed Operating Systems: Concepts and Design”, Pradeep K Sinha, Prentice Hall of India, 2007.
2.	Advanced concepts in operating systems, Mukesh Singhal and Niranjana G. Shivaratri McGraw-Hill, Inc., 1994.
3.	Distributed Systems: Principles and Paradigms, Tanenbaum A.S., Van Steen M., Pearson Education, 2007.
4.	Distributed Computing, Principles and Applications, Liu M.L., Pearson Education, 2004.
5.	Distributed Algorithms, Nancy A Lynch, Morgan Kaufman Publishers, USA, 2003.

Code	Category	L	T	P	C	I.M	E.M	Exam
M19IT1207	PE	3	0	0	3	25	75	3 Hrs.
SOCIAL NETWORK ANALYTICS								
(Program Elective-III)								
Course Objectives:								
1	The learning objective of the course Social Network Analysis is to provide students with essential knowledge of network analysis applicable to real world data, with examples from today's most popular social networks.							
S.No	Outcome							Knowledge Level
1	Identify social network analysis and measures							K3
2	Analyze random graph models and navigate social networks data							K4
3	Apply the network topology and Visualization tools.							K3
4	Analyze the experiment with small world models and clustering models.							K4
5	Compare the application driven virtual communities from social network Structure.							K4
SYLLABUS								
UNIT-I (10 Hrs)	Social Network Analysis: Preliminaries and definitions, Erdos Number Project, Centrality measures, Balance and Homophily.							
UNIT-II (10 Hrs)	Random graph models: Random graphs and alternative models, Models of network growth, Navigation in social Networks, Cohesive subgroups, Multidimensional Scaling, Structural equivalence, roles and positions.							
UNIT-III (10 Hrs)	Network topology and diffusion, Contagion in Networks, Complex contagion, Percolation and information, Navigation in Networks Revisited.							
UNIT-IV (10 Hrs)	Small world experiments, small world models, origins of small world, Heavy tails, Small Diameter, Clustering of connectivity, The ErdosRenyi Model, Clustering Models.							
UNIT-V (10 Hrs)	Network structure -Important vertices and page rank algorithm, towards rational dynamics in networks, basics of game theory, Coloring and consensus, biased voting, network formation games, network structure and equilibrium, behavioral experiments, Spatial and agent-based models.							
Text Books:								
1.	S. Wasserman and K. Faust. Social Network Analysis: Methods and Applications (Cambridge, Cambridge University Press, 1994)							
2.	D. Easley and J. Kleinberg, Networks, Crowds and Markets: Reasoning about a highly connected world							
Reference Books:								
1.	Social Network Data Analytics, Aggarwal, Charu C. (Ed.), Springer Publisher, 2011							

Code	Category	L	T	P	C	I.M	E.M	Exam
M19IT1208	PE	3	0	0	3	25	75	3 Hrs.
DIGITAL IMAGE PROCESSING								
(Program Elective-IV)								
Course Objectives:								
1.	Describe and explain basic principles of digital image processing							
2.	Design and implement algorithms that perform basic image processing (e.g. noise removal and image enhancement).							
3.	Design and implement algorithms for advanced image analysis (e.g. image compression, image segmentation).							
4.	Assess the performance of image processing algorithms and systems.							
Course Outcomes								
S.No	Outcome							Knowledge Level
1	Identify the components of image processing							K3
2	Classify various filtration techniques.							K4
3	Apply image compression techniques.							K3
4	Simplify the concepts of wavelet transforms							K4
5	Analyze the concept of morphological image processing.							K4
SYLLABUS								
UNIT-I (10Hrs)	Introduction: Fundamental steps in Image Processing System, Components of Image Processing System, Elements of Visual Perception, Image Sensing and acquisition, Image sampling & Quantization, Basic Relationship between pixels. Image Enhancement Techniques: Spatial Domain Methods: Basic grey level transformation, Histogram equalization, Image subtraction, image averaging.							
UNIT-II (10Hrs)	Spatial filtering: Smoothing, sharpening filters, Laplacian filters, Frequency domain filters, Smoothing and sharpening filters, Homomorphism is filtering. Image Restoration & Reconstruction: Model of Image Degradation/restoration process, Noise models, Spatial filtering, Inverse filtering, Minimum mean square Error filtering, constrained least square filtering, Geometric mean filter, Image reconstruction from projections. Color Fundamentals, Color Models, Color Transformations.							
UNIT-III (10Hrs)	Image Compression: Redundancies- Coding, Interpixel, Psycho visual; Fidelity, Source and Channel Encoding, Elements of Information Theory; Loss Less and Lossy Compression; Run length coding, Differential encoding, DCT, Vector quantization, Entropy coding, LZW coding; Image Compression Standards-JPEG, JPEG 2000, MPEG; Video compression.							
UNIT-IV (10Hrs)	Wavelet Based Image Compression: Expansion of functions, Multi-resolution analysis, Scaling functions, MRA refinement equation, Wavelet series expansion, Discrete Wavelet Transform (DWT), Continuous, Wavelet Transform, Fast Wavelet Transform, 2-D wavelet							

	Transform, JPEG-2000 encoding.
UNIT-V (10Hrs)	Image Segmentation: Discontinuities, Edge Linking and boundary detection, Thresholding, Region Based Segmentation, Watersheds; Introduction to morphological operations; binary morphology- erosion, dilation, opening and closing operations, applications; basic gray-scale morphology operations; Feature extraction; Classification; Object recognition. Digital Image Watermarking: Introduction, need of Digital Image Watermarking, applications of watermarking in copyright protection and Image quality analysis.
Text Books:	
1.	'Digital Image Processing. 2nd ed. Gonzalez, R.C. and Woods, R.E. India: Person Education,2009
Reference Books:	
1	Digital Image Processing. John Wiley, Pratt, W. K, Fourth Edition-2001
2	Digital Image Processing, Jayaraman, S., Veerakumar, T. and Esakkiranjana, S.,Tata McGraw-Hill, Edition-3,2009

Code	Category	L	T	P	C	I.M	E.M	Exam
M19IT1209	PE	3	0	0	3	25	75	3 Hrs.
BLOCK CHAIN TECHNOLOGIES								
(Program Elective-IV)								
Course Objectives:								
1.	Understand how block chain systems (mainly Bit coin and Ethereum) work,							
2.	To securely interact with them							
3.	Design, build, and deploy smart contracts and distributed applications							
4.	Integrate ideas from block chain technology into their own projects							
Course Outcomes								
S.No	Outcome							Knowledge Level
1	Analyse the foundation of the Block chain technology and understand the processes in payment and funding.							K2
2	Identify the risks involved in building Block chain applications							K3
3	Analyse the legal implications using smart contracts.							K3
4	Choose the present landscape of Blockchain implementations and Understand Crypto currency markets							K3
5	Examine how to profit from trading crypto currencies.							K3
SYLLABUS								
UNIT-I (10Hrs)	The consensus problem ,Asynchronous Byzantine Agreement ,AAP protocol and its analysis ,Nakamoto Consensus on permission-less, nameless, peer-to-peer network - Abstract Models for BLOCKCHAIN - GARAY model ,RLA Model - Proof of Work (PoW) as random oracle - formal treatment of consistency, liveness and fairness - Proof of Stake (PoS) based Chains - Hybrid models (PoW + PoS).							
UNIT-II (10Hrs)	Cryptographic basics for cryptocurrency - a short overview of Hashing, signature schemes, encryption schemes and elliptic curve cryptography.							
UNIT-III (10Hrs)	Bitcoin, Wallet, Blocks, Merkley Tree, hardness of mining,transaction verifiability - anonymity - forks - double spending - mathematical analysis of properties of Bitcoin.							
UNIT-IV (10Hrs)	Ethereum - Ethereum Virtual Machine (EVM), Wallets for Ethereum - Solidity, Smart Contracts, some attacks on smart contracts.							
UNIT-V (10Hrs)	(Trends and Topics) - Zero Knowledge proofs and protocols in Blockchain, Succinct non interactive argument for Knowledge (SNARK), pairing on Elliptic curves, Zcash.							
Text Books:								
1.	Aravind Narayanan, Joseph Bonneau, Edward Felten, Andrew Miller, and Steven Goldfeder. Bitcoin and cryptocurrency technologies: a comprehensive introduction. Princeton University Press, 2016. (Free download available)							

Reference Books:	
1	Joseph Bonneau et al, SoK: Research perspectives and challenges for Bitcoin and cryptocurrency, IEEE Symposium on security and Privacy, 2015 (article available for free download) { curtain raiser kind of generic article, written by seasoned experts and pioneers}.
2	J. A. Garay et al, The bitcoin backbone protocol - analysis and applications EUROCRYPT 2015 LNCS VOL 9057, (VOLII), pp 281-310. (Also available at eprint.iacr.org/2016/1048). (serious beginning of discussions related to formal models for bitcoin protocols).
3	R. Pass et al, Analysis of Blockchain protocol in Asynchronous networks, EUROCRYPT 2017, (eprint.iacr.org/2016/454). A significant progress and consolidation of several principles).

Code	Category	L	T	P	C	I.M	E.M	Exam
M19IT1210	PE	3	0	0	3	25	75	3 Hrs.
DATA SCIENCE								
(Program Elective-IV)								
Course Objectives:								
1.	Provide you with the knowledge and expertise to become a proficient data scientist.							
2.	Demonstrate an understanding of statistics and machine learning concepts that are vital for data science.							
3.	Produce Python code to statistically analyze a dataset.							
4.	Critically evaluate data visualizations based on their design and use for communicating stories from data.							
Course Outcomes By the end of this course student will be able to								
S.No	Outcome							Knowledge Level
1.	Analyse how data is collected, managed and stored for data science.							K4
2.	Identify the key concepts in data science, including their real-world applications and the toolkit used by data scientists.							K3
3.	Apply data collection and management scripts using MongoDB.							K3
SYLLABUS								
UNIT-I (10 Hrs)	Introduction to core concepts and technologies: Introduction, Terminology, data science process, data science toolkit, Types of data, Example applications.							
UNIT-II (10 Hrs)	Data collection and management: Introduction, Sources of data, Data collection and APIs, Exploring and fixing data, Data storage and management, Using multiple data sources							
UNIT-III (10 Hrs)	Data analysis: Introduction, Terminology and concepts, Introduction to statistics, Central tendencies and distributions, Variance, Distribution properties and arithmetic, Samples/CLT, Basic machine learning algorithms, Linear regression, SVM, Naive Bayes.							
UNIT-IV (10 Hrs)	Data visualization-Introduction, Types of data visualization, Data for Visualization- Data types, Data encodings, Retinal variables, Mapping variables to encodings, Visual encodings.							
UNIT-V (10 Hrs)	Applications of Data Science, Technologies for visualization, Bokeh (Python), Recent trends in various data collection and analysis techniques, various visualization techniques, application development methods of used in data science.							
Text Books:								
1.	Cathy O’Neil and Rachel Schutt. Doing Data Science, Straight Talk From The Frontline. O’Reilly.							
2.	Jure Leskovek, Anand Rajaraman and Jeffrey Ullman. Mining of Massive Datasets. v2.1, Cambridge University Press.							

Reference Books:
1. Data Science for dummies, 2nd Edition, Lillian Louise Pierson

Code	Category	L	T	P	C	I.M	E.M	Exam
M19IT1211	PE	3	0	0	3	25	75	3 Hrs.
SOFT COMPUTING								
(Program Elective-IV)								
Course Objectives:								
1.	Develop the skills to gain a basic understanding of neural network theory and fuzzy logic theory.							
2.	Introduce students to artificial neural networks and fuzzy theory from an engineering perspective							
Course Outcomes								
S.No	Outcome							Knowledge Level
1	Classify the fuzzy logic and the concept of fuzziness involved in various systems and fuzzy set theory.							K4
2	Simplify the concepts of fuzzy sets, knowledge representation using fuzzy rules, approximate reasoning, fuzzy inference systems, and fuzzy logic							K4
3	Identify fundamental theory and concepts of neural networks, Identify different neural network architectures, algorithms, applications and their limitations							K3
4	Identify appropriate learning rules for each of the architectures and learn several neural network paradigms and its applications							K3
5	Analyze different applications of these models to solve engineering and other problems..							K4
SYLLABUS								
UNIT-I (10Hrs)	Introduction to Neuro – Fuzzy and Soft Computing, Fuzzy Sets, Basic Definition and Terminology, Set-theoretic Operations, Member Function Formulation and Parameterization, Fuzzy Rules and Fuzzy Reasoning, Extension Principle and Fuzzy Relations, Fuzzy If-Then Rules, Fuzzy Reasoning, Fuzzy Inference Systems, Mamdani Fuzzy Models, Surgeon Fuzzy Models, Tsukamoto Fuzzy Models, Input Space Partitioning and Fuzzy Modeling.							
UNIT-II (10Hrs)	Derivative based Optimization, Descent Methods, The Method of Steepest Descent, Classical Newton’s Method, Step Size Determination, Derivative-free Optimization, Genetic Algorithms, Simulated Annealing and Random Search – Downhill Simplex Search.							
UNIT-III (10Hrs)	Introduction, Knowledge Representation, Reasoning, Issues and Acquisition: Propositional and Predicate Calculus Rule Based knowledge Representation Symbolic Reasoning under Uncertainty Basic knowledge Representation Issues Knowledge acquisition, Heuristic Search: Techniques for Heuristic search Heuristic Classification State Space Search: Strategies Implementation of Graph Search based on Recursion Patent directed Search Production System and Learning.							
UNIT-IV (10Hrs)	Adaptive Neuro-Fuzzy Inference Systems, Architecture – Hybrid Learning Algorithm, Learning Methods that Cross-fertilize ANFIS and RBFN – Coactive Neuro Fuzzy Modeling, Framework Neuron Functions for Adaptive Networks – Neuro Fuzzy Spectrum.							

UNIT-V (10Hrs)	Printed Character Recognition, Inverse Kinematics Problems, Automobile Fuel Efficiency Prediction, Soft Computing for Color Recipe Prediction.
Text Books:	
1.	J.S.R.Jang, C.T.Sun and E.Mizutani, “Neuro-Fuzzy and Soft Computing”, PHI, 2004, Pearson Education 2004.
2	N.P.Padhy, “Artificial Intelligence and Intelligent Systems”, Oxford University Press, 2006.
Reference Books:	
1	Elaine Rich & Kevin Knight, Artificial Intelligence, Second Edition, Tata Mcgraw Hill Publishing Comp., 2006, New Delhi.

Code	Category	L	T	P	C	I.M	E.M	Exam
M19IT1212	PE	3	0	0	3	25	75	3 Hrs.
NATURAL LANGUAGE PROCESSING								
(Program Elective-IV)								
Course Objectives:								
1.	Make them understand the concepts of morphology, syntax, semantics and pragmatics of the language and that they are able to give the appropriate examples that will illustrate the above mentioned concepts.							
2.	Teach them to recognize the significance of pragmatics for natural language understanding.							
3	Enable students to be capable to describe the application based on natural language processing and to show the points of syntactic, semantic and pragmatic processing.							
Course Outcomes								
S.No	Outcome							Knowledge Level
1	Identify approaches to syntax and semantics in NLP							K3
2	Classify approaches to discourse, generation, dialogue and summarization within NLP.							K4
3	Identify current methods for statistical approaches to machine translation.							K3
4	Identify machine learning techniques used in NLP, including hidden Markov models and probabilistic							K3
5	Distinguish context-free grammars, clustering and unsupervised methods, log-linear and discriminative models, and the EM algorithm as applied within NLP							K4
SYLLABUS								
UNIT-I (10Hrs)	Introduction: NLP tasks in syntax, semantics, and pragmatics. Applications such as information extraction, question answering, and machine translation. The problem of ambiguity. The role of machine learning. Brief history of the field.							
UNIT-II (10Hrs)	N-gram Language Models: The role of language models, Simple Ngram models. Estimating parameters and smoothing. Evaluating language models. Part of Speech Tagging and Sequence Labeling: Lexical syntax. Hidden Markov Models. Maximum Entropy Models. Conditional Random Fields.							
UNIT-III (10Hrs)	Syntactic parsing: Grammar formalisms and tree banks. Efficient parsing for context-free grammars (CFGs). Statistical parsing and probabilistic CFGs (PCFGs). Lexicalized PCFGs.							
UNIT-IV (10Hrs)	Semantic Analysis: Lexical semantics and word-sense disambiguation. Compositional semantics. Semantic Role Labeling and Semantic Parsing.							
UNIT-V (10Hrs)	Information Extraction (IE) and Machine Translation (MT): Named entity recognition and relation extraction. IE using sequence labeling. Basic issues in MT. Statistical translation, word alignment, phrase based translation, and synchronous							

	grammars. Dialogues: Turns and utterances, grounding, dialogue acts and structures Natural Language Generation: Introduction to language generation, architecture, discourse planning (text schemata, rhetorical relations).
Text Books:	
1.	D. Jurafsky & J. H. Martin – “Speech and Language Processing – An introduction to Language processing, Computational Linguistics, and Speech Recognition”, Pearson Education
Reference Books:	
1.	Allen, James. 1995. – “Natural Language Understanding”. Benjamin/ Cummings, 2ed.
2.	Bharathi, A., Vineet Chaitanya and Rajeev Sangal. 1995. Natural Language Processing- “A Pananian Perspective”. Prentice Hll India, Eastern Economy Edition.

Code	Category	L	T	P	C	I.M	E.M	Exam
M19IT1213	PC	0	0	4	2	25	75	3 Hrs.
ADVANCE ALGORITHMS LAB								
Course Objectives:								
To Understand								
1.	Knowing about oops concepts for a specific problem							
2.	Various advanced data structures concepts like arrays, stacks, queues, linked lists, graphs and trees.							
3.	Principles for good program design, especially the uses of data strucure							
4.	Various aspects of algorithm development							
5.	Qualities of a good solution							
Course Outcomes								
S.No	Outcome							Knowledge Level
1.	Identify different classes of problems concerning their computation difficulties							K3
2.	Analyze the performance of the Advanced Algorithm							K4
3.	Apply advanced algorithm techniques to solve real world problems.							K4
LIST OF EXPERIMENTS								
1.	Write a program to implement topological sorting.							
2.	Write a program to implement Quick Sort of given elements							
3.	Write a program to Implement DFS for a Graph.							
4.	Write a program to implement BFS for a Graph							
5.	Write a program Implement Krushkal’s algorithm to generate a min-cost spanning tree							
6.	Write a program Implement Prim’s algorithm to generate a min-cost spanning tree.							
7.	Write a program to implement dijkstra's greedy algorithm for single source shortest path problem							
8.	Write a program to implement Ford-Fulkerson algorithm for computing a maximum flow in a network							
9.	Write a program to implement Dynamic Programming algorithm for the chained matrix multiplication							
10.	Write a program to implement Floyd's dynamic programming algorithm for the all pair shortest path problem							
11.	Write a program to implement Approximate algorithm for Vertex-Cover Problem							
12.	Write a program to implement Approximate algorithm solve Travelling Salesperson Problem							
Reference Books:								
1.	Algorithm Design, J. Kleinberg and E.Tardos, Pearson(Addison Wesley)							
2.	Computer Algorithms, Introduction to design and analysis, 3rd Edition , Sara Baase, Allen, Van, Gelder, Pearson Education							

Code	Category	L	T	P	C	I.M	E.M	Exam
M19IT1214	PC	0	0	4	2	25	75	3 Hrs.
FULL STACK TECHNOLOGIES LAB								
Course Objectives:								
1.	Translate user requirements into the overall architecture and implementation of new systems and Manage Project and coordinate with the Client.							
2.	Write backend code in Python/Java, PHP languages and Writing optimized front end code HTML and JavaScript.							
3.	Understand, create and debug database related queries and Create test code to validate the applications against client requirement.							
4.	Monitor the performance of web applications & infrastructure and Troubleshooting web application with a fast and accurate a resolution.							
Course Outcomes								
S.No	Outcome							Knowledge Level
1.	Identify the Basic Concepts of Web & Markup Languages							K3
2.	Develop web Applications using Scripting Languages & Frameworks							K3
3.	Creating & Running Applications using JSP libraries							K4
4.	Creating Our First Controller Working with and Displaying in Angular Js and Nested Forms with ng-form							K3
5.	Creating & Running Back-end scripts & Connecting to Databases							K4
Artificial Intelligence Specialization :								
1.	Implementation of ‘get’ and ‘post’ methods.							
2.	CSS implementation in colors, boarder padding.							
3.	CSS3 implementation button frames tables, navigation bars.							
4.	Create registration and login forms with validations using Jscript query.							
5.	Jscript to retrieve student information from student database using database connectivity.							
6.	Angular Js data binding							
7	Angular JS directives and Events							
8	Using angular Js fetching data from MySQL.							
9	Write a PHP program for registering users of a website and login.							
10	User Authentication: Assume four users user1, user2, user3 and user4 having the passwords pwd1, pwd2, pwd3 and pwd4 respectively. Write a PHP for doing the following. Write a program to create a Cookie and add these four user id’s and passwords to this Cookie. b. Read the user id and passwords entered in the Login form (exp1) and authenticate with the values (user id and passwords) available in the cookies. If he is a valid user (i.e., user-name and password match) you should welcome him by name (user-name) else you should display “You are not an authenticated user ”. Use init-parameters to do this.							
11	Install a database (Mysql or Oracle): Create a table which should contain at least the following fields: name, password, email-id, phone number (these should hold the data from the registration form).							

	a) Write a PHP program to connect to that database and extract data from the tables and display them. b) Experiment with various SQL queries. c) Insert the details of the users who register with the web site, whenever a new user clicks the submit button in the registration page.
12	Write a PHP program which does the following job: Insert the details of the 3 or 4 users who register with the web site by using registration form. Authenticate the user when he submits the login form using the user name and password from the database (Similar to week8 instead of cookies).
Reference Books:	
1.	Jeffrey C. Jackson, "Web Technologies--A Computer Science Perspective", Pearson Education, 2006
2.	Robert. W. Sebesta, "Programming the World Wide Web", Fourth Edition, Pearson Education, 2007
3.	AngularJS: Up and Running Enhanced Productivity with Structured Web Apps By Brad Green, Shyam Seshadri Publisher: O'Reilly Media
4.	Learning React Functional Web Development with React and Redux By Alex Banks, Eve Porcello Publisher: O'Reilly Media
5.	Head First Java, 2nd Edition by Bert Bates, Kathy Sierra Publisher: O'Reilly Media, Inc

Code	Category	L	T	P	C	I.M	E.M	Exam
M19IT1215	MP	0	0	4	2	100	--	3 Hrs.
MINI PROJECT WITH SEMINAR								
For Mini Project with Seminar, a student under the supervision of a faculty member, shall collect the literature on a topic and critically review the literature and submit it to the department in a report form and shall make an oral presentation before the Project Review Committee consisting of Head of the Department, supervisor/mentor and two other senior faculty members of the department. For Mini Project with Seminar, there will be only internal evaluation of 100 marks. A candidate has to secure a minimum of 50% of marks to be declared successful.								

Code	Category	L	T	P	C	I.M	E.M	Exam
#AC-2	AC	2	0	0	0	0	0	--
AUDIT COURSE-2								
List of Audit Courses and their Syllabi are mentioned in the First Semester Syllabus. The students can opt any one course for AC 2 from the list mentioned in first semester by not opting the course which is already taken for AC 1								



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CHINNA AMIRAM (P.O):: BHIMAVARAM :: W.G.Dt., A.P., INDIA :: PIN: 534 204

SCHEME OF INSTRUCTION & EXAMINATION (Regulation R19) **M.TECH (INFORMATION TECHNOLOGY)** **DEPARTMENT OF INFORMATION TECHNOLOGY** (With effect from 2019-2020 Admitted Batch onwards)

III-SEMESTER

Subject Code.	Name of the Subject	Category	C	L	T	P	Internal Marks	External Marks	Total Marks
#PE-V/ MOOCS	Program Elective-V	PE	3	3	0	0	25	75	100
#OE-I/ MOOCS	Open Elective-I	OE	3	3	0	0	25	75	100
M19IT2107	Dissertation-I /Industrial Project#	PR	10	0	0	20	50	50	100
TOTAL			16	6	0	20	100	200	300

	Code	Course
#PE-V/ MOOCS	M19IT2101	Deep Learning
	M19IT2102	Embedded Computing
	M19IT2103	Ethical Hacking
	M19IT2104	Digital Marketing
	M19IT2105 (MOOCS-I)	Students Going for Industrial Project / Thesis will complete these courses through MOOCS. Students can also choose SWAYAM or NPTEL with a 12 weeks' course duration in PG level with 3 credits, but the chosen subject should not be covered in their M. Tech Course
#OE-I/ MOOCS	#OE-I	Students have to choose one open elective course offered by departments other than the IT & CSE departments. List of open Electives offered by other departments are enclosed.
	M19IT2106 (MOOCS-II)	Students Going for Industrial Project / Thesis will complete these courses through MOOCS. Students can also choose SWAYAM or NPTEL with a 12 weeks' course duration in PG level with 3 credits, but the chosen subject should not be covered in their M. Tech Course

OPEN ELECTIVES OFFERED TO OTHER DEPARTMENTS

Code	Course
M19IT2108	Web Technologies
M19IT2109	Internet of Things
M19IT2110	Machine Learning

Code	Category	L	T	P	C	I.M	E.M	Exam
M19IT2101	PE	3	0	0	3	25	75	3 Hrs.
DEEP LEARNING								
(Program Elective-V)								
Course Objectives:								
1.	Learn deep learning methods for working with sequential data							
2.	Learn deep recurrent and memory networks							
3.	Apply such deep learning mechanisms to various learning problems							
4.	Learn deep Turing machines, the open issues in deep learning, and have a grasp of the current research directions.							
Course Outcomes								
After completion of course, students would be able to:								
S.No	Outcome							Knowledge Level
1.	Demonstrate the basic concepts fundamental learning techniques and layers.							K2
2.	Discuss the Neural Network training, various random models.							K3
3.	Explain different types of deep learning network models.							K2
4.	Classify the Probabilistic Neural Networks.							K3
5.	Implement tools on Deep Learning techniques.							K4
SYLLABUS								
UNIT-I (10 Hrs)	Introduction: Various paradigms of learning problems, Perspectives and Issues in deep learning framework, review of fundamental learning techniques. Feed forward neural network: Artificial Neural Network, activation function, multi-layer neural network							
UNIT-II (10 Hrs)	Training Neural Network: Risk minimization, loss function, back propagation, regularization, model selection, and optimization. Deep Neural Networks: Difficulty of training deep neural networks, Greedy layer wise training.							
UNIT-III (10 Hrs)	Deep Learning: Deep Feed Forward network, regularizations, training deep models, dropouts, Convolution Neural Network, Recurrent Neural Network, and Deep Belief Network.							
UNIT-IV (10 Hrs)	Probabilistic Neural Network: Hopfield Net, Boltzmann machine, RBMs, Sigmoid net, Auto encoders.							
UNIT-V (10 Hrs)	Applications: Object recognition, sparse coding, computer vision, natural language processing. Introduction to Deep Learning Tools: Tensor Flow, Caffe, Theano, Torch.							

Text Books:	
1.	Goodfellow, I., Bengio, Y., and Courville, A., Deep Learning, MIT Press, 2016
2.	Bishop, C. ,M., Pattern Recognition and Machine Learning, Springer, 2006.
Reference Books:	
1.	Yegnanarayana, B., Artificial Neural Networks PHI Learning Pvt. Ltd, 2009.
2.	Golub, G.,H., and Van Loan,C.,F., Matrix Computations, JHU Press,2013.
3.	Satish Kumar, Neural Networks: A Classroom Approach, Tata McGraw-Hill Education, 2004
4.	Neural Networks: A Systematic Introduction, Raúl Rojas, 1996
5.	Pattern Recognition and Machine Learning, Christopher Bishop, 2007

Code	Category	L	T	P	C	I.M	E.M	Exam
M19IT2102	PE	3	0	0	3	25	75	3 Hrs.
EMBEDDED COMPUTING								
(Program Elective-V)								
Course Objectives:								
1.	To demonstrate the basic functions of the Linux and embedded system							
2.	To demonstrate about Linux Kernel							
3.	To demonstrate about I/O and File Systems							
4.	To develop client server models using TCP socket programming							
Course Outcomes								
After completion of course, students would be able to:								
S.No	Outcome							Knowledge Level
1.	Knowledge and understanding of Embedded Linux OS Architecture, Linux Kernel Modules							K2
2.	Describes the differences between the general computing system and the embedded computing system.							K3
3.	Write client server program using TCP sockets							K3
SYLLABUS								
UNIT-I (10 Hrs)	Linux and Embedded Systems: An Introduction- What is an Embedded System?, Embedded system components, Basic software, Operating systems for embedded systems, Why Linux- based embedded systems?, Linux evolution, Linux-based Embedded System Component Stack- Linux-based embedded system components, Reference hardware model, Reference hardware model implementations, CPU memory map, The role of the bootloader, Possible scenarios, An example of bootloader operations.							
UNIT-II (10 Hrs)	Linux kernel, Device tree, System programs, Application, Typical layout of the root filesystem, Anatomy of a Linux-based System - Linux architecture, Conceptual view of the kernel, Process scheduler, Memory manager, Memory manager external interfaces, Memory manager architecture, Virtual file system, i-node, i-node interface, File interface, Virtual file system architecture, Inter-process communication, Inter-process communication architecture							
UNIT-III (10 Hrs)	Introduction to Linux kernel modules- Introduction, CPU – I/O interface, CPU – I/O interface with polling, CPU – I/O interface with interrupt, CPU – I/O interface, CPU – I/O interface latency, Direct memory access (DMA) architecture, DMA transfer modes, I/O taxonomy, Typical operations, Linux devices, The Virtual File System (VFS) abstraction, VFS– an example, VFS functions– include/linux/fs.h, The device file concept, Linux kernel modules – the initialization function, the cdev data structure, the initialization function, the clean-up function, custom VFS functions							
UNIT-IV (10 Hrs)	File Handling: Memory Mapping- Page Alignment, Establishing Memory Mappings, Unmapping Regions, Syncing Memory Regions to Disk, Locking Memory Regions, File							

	Locking, Lock Files, Record Locking, Mandatory Locks, Leasing a File.
UNIT-V (10 Hrs)	Networking with Sockets: Protocol Support, Nice Networking, Real Networking, Making Reality Play Nice, Addresses, Utility Functions, Basic Socket Operations, Creating a Socket, Establishing Connections, Binding an Address to a Socket, Waiting for Connections, Connecting to a Server, Finding Connection Addresses, Networking Machines with TCP/IP, Byte Ordering, IPv4 Addressing, IPv6 Addressing, Manipulating IP Addresses, Turning Names into Addresses, Turning Addresses into Names, Listening for TCP Connection, TCP Client Applications
Text Books:	
1.	Embedded Linux Systems with the Yocto Project by Rudolf K. Sterif
2.	Linux Application Development, Michael K. Johnson, Erik W. Troan, O' Reilly, 2004
Reference Books:	
1.	Peter Barry and Patrick Crowley, "Modern Embedded Computing", 1st Edition., Elsevier/Morgan Kaufmann, 2012.

Code	Category	L	T	P	C	I.M	E.M	Exam
M19IT2103	PE	3	0	0	3	25	75	3 Hrs.
ETHICAL HACKING								
(Program Elective-V)								
Course Objectives:								
1.	To gain knowledge about Ethical hacking and penetration testing.							
2.	To learn about various types of attacks, attackers and security threats and vulnerabilities present in the computer system.							
3.	To examine how social engineering can be done by attacker to gain access of useful & sensitive information about the confidential data.							
4.	To learn about cryptography, and basics of web application attacks.							
5.	To gain knowledge of the tools , techniques and ethical issues likely to face the domain of ethical hacking and ethical responsibilities.							
Course Outcomes								
After completion of course, students would be able to:								
S.No	Outcome							Knowledge Level
CO1	Remember various hacking methods, system security vulnerability testing.							K1
CO2	Apply system vulnerability attacks and demonstrate a security assessment report							K3
CO3	Understand various issues related to hacking							K2
SYLLABUS								
UNIT-I (10 Hrs)	Hacking Windows: BIOS Passwords, Windows Login Passwords, Changing Windows Visuals, Cleaning Your Tracks, Internet Explorer Users, Cookies, URL Address Bar, Netscape Communicator, Cookies URL History, The Registry, Baby Sitter Programs.							
UNIT-II (10 Hrs)	Advanced Windows Hacking: Editing your Operating Systems by editing Explorer.exe, The Registry, The Registry Editor, Description of .reg file, Command Line Registry Arguments, Other System Files, Some Windows & DOS Tricks, Customize DOS, Clearing the CMOS without opening your PC, The Untold Windows Tips and Tricks Manual, Exiting Windows the Cool and Quick Way, Ban Shutdowns: A Trick to Play, Disabling Display of Drives in My Computer, Take Over the Screen Saver, Pop a Banner each time Windows Boots, Change the Default Locations, Secure your Desktop Icons and Settings.							
UNIT-III (10 Hrs)	Getting Past the Password: Passwords: An Introduction, Password Cracking, Cracking the Windows Login Password, The Glide Code, Windows Screen Saver Password, XOR, Internet Connection Password, Sam Attacks, Cracking Unix Password Files, HTTP Basic Authentication, BIOS Passwords, Cracking Other Passwords.							
UNIT-IV (10 Hrs)	The Perl Manual: Perl: The Basics, Scalars, Interacting with User by getting Input, Chomp() and Chop(), Operators, Binary Arithmetic Operators, The Exponentiation Operator(**), The Unary Arithmetic Operators, Other General Operators, Conditional Statements, Assignment Operators. The : Operator, Loops, The While Loop, The For Loop, Arrays, THE FOR EACH LOOP: Moving through an Array, Functions Associated with Arrays, Push() and Pop(), Unshift() and Shift(), Ssplice(), Default Variables, \$_, @ARGV, Input Output,							

	Opening Files for Reading, Another Special Variables.
UNIT-V (10 Hrs)	Virus Working, Boot Sector Viruses (MBR or Master Boot Record), File or Program Viruses, Multipartite Viruses, Stealth Viruses, Polymorphic Viruses, Macro Viruses, Blocking Direct Disk Access, Recognizing Master Boot Record (MBR) Modifications, Identifying Unknown Device Drivers, making own Virus, Macro Viruses, Using Assembly to Create your own Virus, Modifying a Virus so Scan won't Catch it, Creating New Virus Strains, Simple Encryption Methods.
Text Books:	
1.	Patrick Engbreton: "The Basics of Hacking and Penetration Testing: Ethical Hacking and Penetration Testing Made Easy", 1st Edition, Syngress publication, 2011.
2.	Ankit Fadia : "Unofficial Guide to Ethical Hacking", 3rd Edition , McMillan India Ltd, 2006.
Reference Books:	
1.	Simpson/backman/corley, "HandsOn Ethical Hacking & Network Defense International", 2nd Edition, Cengageint, 2011.

Code	Category	L	T	P	C	I.M	E.M	Exam
M19IT2104	PE	3	0	0	3	25	75	3 Hrs.
DIGITAL MARKETING								
(Program Elective-V)								
Course Objectives:								
1.	Digital marketing aims at being SMART (Specific, Measurable, Achievable, Relevant and Time Related) so that people can withstand against competitors.							
Course Outcomes								
After completion of course, students would be able to:								
S.No	Outcome							Knowledge Level
1.	Explain about web pages with basic HTML5, DHTML tags using CSS and XML, the overview of W3C DOM.							K2
2.	Student can apply digital Java Scripts to applications							K3
3.	Apply search engine optimization techniques to a website.							K4
4.	Illustrate how the effectiveness of a digital marketing campaign can be measured							K4
5.	Apply advanced practical skills in common digital marketing tools such as SEO, SEM, Social media and Blogs							K4
SYLLABUS								
UNIT-I (10 Hrs)	HTML: Introduction, HTML5, Audio Elements, Video Elements, Organizing Elements. Scripting Documents: Dynamic Document content, Document properties, Legacy DOM, Document Collections, Overview of the W3C DOM, Traversing a Document, Finding Elements in a Document, Modifying a Document, Adding Content to a Document Example							
UNIT-II (10 Hrs)	Cascading Style Sheets and Dynamic HTML: Overview of CSS, CSS for DHTML Scripting inline Styles, Scripting computed styles, Scripting CSS Classes, Scripting Style Sheets, Java Script and XML: Obtaining XML Documents, Manipulating XML with the DOM API, Transforming XML with XSLT querying XML with X path, Serializing XML, Example, XML and Web services.							
UNIT-III (10 Hrs)	Search Engine Optimization (SEO): Searching Engine Marketing, Search Engine Optimization, Measuring SEO Success, Mapping with SEO Journey, Search Advertising: Online Advertising Payment Models, Search Advertising (Desktop & Mobile Devices), Planning & Executing a search Advertising Campaign, Strategic Implications of Advertising on the search Network.							
UNIT-IV (10 Hrs)	Search Media Marketing: What is Social Media? Social Media Marketing, Social Media Marketing Strategy, Adopting Social Media in Organizations: Internal Learning, Paid-Owned-Earned Media, Social CRM, Mobile Marketing: Mobile Internet in India, What is Mobile Marketing? Email Marketing Strategy, Forms of Mobile Marketing, Mobile Advertising, M-Commerce.							

UNIT-V (10 Hrs)	E-Mail Marketing: E-Mail Marketing in India, What is E-Mail Marketing? E-Mail Marketing Strategy, Executing E-Mail Marketing, Internet Marketing: Internet Marketing Strategy, Content Marketing, Content Marketing in India.
Text Books:	
1.	The Art of Digital Marketing: The Definitive Guide to Creating Strategic, Targeted, and Measurable Online Campaigns, Ian Dodson, Wiley, 2016
2.	Programming the World Wide Web, Robert W Sebesta, Pearson, 8th edition, 2015
Reference Books:	
1.	Fundamentals of Digital Marketing, Second Edition, Pearson Paperback, 2019
2.	Internet Marketing- A Practical approach in the India Context by Moutusy Maity, Oxford
3.	Java Script: The Definite Guide David Flanagan, O' Reilly Publisher

Code	Category	L	T	P	C	I.M	E.M	Exam
M19IT2105	PE	0	0	0	3	25	75	3 Hrs.
MOOCS-I								
(Program Elective-V)								
Students Going for Industrial Project / Thesis will complete these courses through MOOCS. Students can also choose SWAYAM or NPTEL with a 12 weeks' course duration in PG level with 3 credits, but the chosen subject should not be covered in their M. Tech Course								

Code	Category	L	T	P	C	I.M	E.M	Exam
	OE	--	--	--	3	25	75	3 Hrs.
OPEN ELECTIVE								
Students have to choose one open elective course offered by departments other than the IT & CSE departments. List of open Electives offered by other departments are given below.								

Offered from	Course Code	Course Name	Offered to
CIVIL ENGINEERING	M19 ST 2107	Construction Management	CST, CS, PSA,IT & CAD/CAM
	M19 ST 2108	Green Technology	
	M19 ST 2109	Analysis of Offshore Structures	
ELECTRONICS & COMMUNICATION ENGINEERING	M19 CS 2107	Signals and systems	ST, CST, PSA, IT & CAD/CAM
	M19 CS 2108	Principles of Communication	
	M19 CS 2109	Image and video Processing	
ELECTRICAL & ELECTRONICS ENGINEERING	M19PS2107	Electric And Hybrid Vehicles	ST, CST, CS, IT & CAD/CAM
	M19PS2108	Energy From Waste	
	M19PS2109	Energy Management and Auditing	
MECHNAICAL ENGINEERING	M19CAD 2107	Operations Research	ST, CST, CS, PSA & IT
	M19CAD 2108	Nano Technology	
	M19CAD 2109	Product Design & Manufacturing	
SCIENCE & HUMANITIES	M19BS2101	Management and Organisational Behaviour	ST, CST, CS, PSA, IT & CAD/CAM

Code	Category	L	T	P	C	I.M	E.M	Exam
M19IT2106	MOOCS	--	--	--	3	25	75	3 Hrs.
MOOCS-II								
Students Going for Industrial Project / Thesis will complete these courses through MOOCS. Students can also choose SWAYAM or NPTEL with a 12 weeks' course duration in PG level with 3 credits, but the chosen subject should not be covered in their M. Tech Course								

Code	Category	L	T	P	C	I.M	E.M	Exam
M19IT2107	PR	0	0	20	10	50	50	3 Hrs.
DISSERTATION-I/INDUSTRIAL PROJECT								
The Student has to register for Dissertation-I / Industrial project in III semester. Student has to submit, in consultation with his project supervisor, the title, objective and plan of action of his project work for approval. The student can initiate the Project work, only after obtaining the approval from the Project Review Committee (PRC). Continuous assessment of Dissertation-I during the III-Semester will be monitored by the PRC. Dissertation-I/ Industrial Project is evaluated for 50 internal marks and 50 external marks. Internal marks 50 awarded by Project Guide and PRC jointly based on continuous assessment consisting of two seminars based on Dissertation work-I. External marks 50 awarded by External Examiner, Supervisor and Head of the Department jointly based on a review and Viva voce on Dissertation work-I.								

SCHEME OF INSTRUCTION & EXAMINATION
 (Regulation R19)
M.TECH (INFORMATION TECHNOLOGY)
DEPARTMNT OF INFORMATION TECHNOLOGY
 (With effect from **2019-2020**Admitted Batch onwards)

IV-SEMESTER

Subject Code.	Name of the Subject	Category.	C	L	T	P	Int. Marks	Ext. Marks
M19IT2201	Dissertation-II /Industrial Project#	PR	16	0	0	32	--	100

Code	Category	L	T	P	C	I.M	E.M	Exam
M19IT2201	PR	0	0	32	16	--	100	3 Hrs.
DISSERTATION-II/INDUSTRIAL PROJECT								
The student has to continue his/her work from Dissertation-I / Industrial project to complete Dissertation-II in IV semester. Continuous assessment of Dissertation-II during IV-Semester will be monitored by the PRC. Dissertation-II is evaluated for 100 external marks based on Review and Viva Voce. Review and Viva-Voce examination shall be conducted by a board consisting of the Supervisor, Head of the Department and the examiner who adjudicated the Thesis. The Board shall jointly report the candidate's work for 100 marks. If the report of the Viva-Voce is unsatisfactory (ie, < 50 marks), the candidate shall retake the Viva-Voce examination only after three months. If he fails to get a satisfactory report at the second Viva-Voce examination, the candidate has to reregister for the project and complete the project within the stipulated time after taking the approval from the College.								